INCH-POUND

MIL-M-29515(EC)
29 April 1988

# MILITARY SPECIFICATION

## MODEM, DIGITAL DATA, MD-1030A(V)

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the performance, design, test, manufacturing and acceptance requirements of the Digital Data Modem, MD-1030A(V), hereinafter referred to as the modem.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A

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SPECICIFICATIONS

MILITARY

| | |
|---|---|
| MIL-S-901 | Shock Tests, H.I.(High Impact); Shipboard Machinery, Equipment And Systems, Requirements For |
| MIF-F-14072 | Finishes For Ground Equipment |
| MIL-E-16400G | Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of |
| MIL-P-55110 | Printed Wiring Boards, General Specification For |

STANDARDS

FERERAL

| | |
|---|---|
| FED-STD-151 | Metals: Test Methods |
| FED-STD-595 | Colors |

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-167-1 | Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited) |
| MIL-STD-188-114 | Electrical Characteristics Of Digital Interface Circuits |
| MIL-STD-275 | Printed Wiring For Electronic Equipment |
| MIL-STD-454 | Standard General Requirements For Electronic Equipment |
| MIL-STD-461 | Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-471 | Maintainability Verification/ Demonstration/Evaluation |
| MIL-STD-781C | Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods And Engineering Guidelines |



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| | |
|---|---|
| MIL-STD-1399, Section 070, Part 1 | Interface Standard For Shipboard Systems, D.C. Magnetic Field Environment (Metric) |
| DoD-STD-1399, Section 300 | Interface Standard For Shipboard Systems, Electric Power Alternating Current (Metric) |
| MIL-STD-1472 | Human Engineering Design Criteria For Military Systems, Equipment And Facilities |
| DoD-STD-1686 | Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric) |

HANDBOOK

MILITARY

| | |
|---|---|
| MIL-HDBK-241 | Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies |

2.1.2 Government drawings and publications. The following Government drawing and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

MILITARY

NAVAL RESEARCH LABORATORY (NRL)

| | |
|---|---|
| NRL 7259302-112 | Schematic, Unit Interface Card, AN/WSC-6(V) |

PUBLICATIONS

MILITARY

ASSISTANT SECRETARY OF THE NAVY (SUPPLY AND LOGISTICS)

| | |
|---|---|
| NAVMAT P 4855-1 | Navy Power Supply Reliability Design And Manufacturing Guidelines |



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NAVAL AIR SYSTEMS COMMAND (NAVAIR)

| | |
|---|---|
| WS-6536E with SCN 1, SCN 2 and SCN 3 | Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections |

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DRAWING

HARRIS CORPORATION

628601-G01 Digital Data Modem, MD-1030A(V)

(Copies of this drawing may be obtained from the contracting officer.)

PUBLICATIONS

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

| | |
|---|---|
| EIA RS-232-C-69 | Interface Between Data Terminal Equipment And Data Communication Equipment Employing Serial Binary Data Interchange |
| EIA RS-310-C-77 | Racks, Panels And Associated Equipment |

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of



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this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The modem shall be in accordance with MIL-E-16400G, to the extent specified herein. The modem will be used to provide the principal communications link through the Defense Satellite Communication Systems series satellites. The identical modem will also be used at the shore terminals for reception and to provide the return link to the ships. The modems are full duplex and will interface between the baseband data signal and the nominal 70-megahertz (MHz) biphase modulated intermediate frequency (IF).

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Modem description. The modem, is a full duplex unit containing four major independent functional elements as specified in a through d:

a. A biphase phase shift keying (PSK) modulator
b. A biphase PSK demodulator
c. An error correcting coder/decoder
d. A multiplexer/demultiplexer (MUX/DEMUX)

The modem shall provide the near-theoretical performance required for use in shipboard or fixed-ground satellite communications terminals. Selection of any three-digit data rate between 75.0 bits per second (bps) and 100 kilobits per second (kbps) shall be provided; when one-half rate encoding is used the data rate will be between 75 bps and 50 kbps. (The symbol rate is twice the data rate when one-half rate encoded.) Performance of this modem is limited to a maximum rate of 65 kbps (32.5 kbps with one-half rate encoding) due to modulator bandlimiting. The modem shall be configured as a mainframe which performs the essential communications functions. The operating signal and performance characteristics for the modem (mainframe) are specified in 3.2 and the control and self-test functions are specified in 3.6.2.3. The modem shall be a completely self-contained unit with built-in power supplies. Plug-in modular construction shall be used throughout to provide ready access for troubleshooting and repair. Modem internal electronics shall be composed primarily of digital integrated circuits (ICs). Operation and maintenance of the modem shall require no internal adjustments.

3.1.2.1 PSK modulator. The PSK modulator shall accept a MIL-STD-188-114-compatible digital data input from the user and shall provide a 70-MHz nominal PSK-modulated output signal to the communications terminal. A transmit bit synchronizer shall be included in the modem and shall provide the capability to operate either of the internally-selectable input modes specified in a and b (MUX Bypass Mode only):



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a. Mode 1: Synchronous mode operation with data input or data and clock input, where the modem synchronizes to the data or clock if the clock is provided

b. Mode 2: Synthesized mode, operation with the modem as master clock source

Switching from Mode 1 to Mode 2 shall be accomplished by a single internal switch. The synchronizer shall provide input jitter smoothing and a twice-rate clock to the internal convolutional encoder. A differential encoder shall be included. The final IF output shall be filtered to reduce adjacent channel interference.

3.1.2.2 PSK demodulator. The PSK demodulator shall accept a noise-corrupted 70-MHz nominal PSK-modulated signal from the communications terminal, and provide a MIL-STD-188-114-compatible recovered data and clock to the user. The PSK demodulator shall contain a carrier regeneration and demodulation circuit, and a bit synchronizer which provide both regenerated clock and soft decision data outputs to the internal maximum-likelihood decoder. A differential decoder shall be included to resolve the one or zero ambiguity at the final output. Output data shall be maintained at a continuous mark level when the demodulator is out-of-lock. Rear panel test signals shall be provided to observe the output data and clock. These test signals shall represent the composite DEMUX input with the data maintained at a high mark level when the demodulator is out-of-lock.

3.1.2.3 Coder/decoder. A one-half rate constraint length 7, convolution encoder and maximum likelihood decoder shall be included to provide processing gain. The convolutional encoder and maximum likelihood decoder functions may be bypassed by internal switch selection.

3.1.2.4 MUX/DEMUX. The MUX/DEMUX is full duplex containing two major functional elements: a time division multiplexer (TDM) and a time division DEMUX. The MUX/DEMUX may be bypassed for operation with a single data channel by front panel switch selection.

3.1.2.4.1 MUX. The MUX shall accept asynchronous or isochronous data from two teletype (TTY) channels, each at an input rate of 75 bps, plus synchronous data at 75 bps, 32 kbps, or 64 kbps. The channels shall be within $\pm 10^{-4}$ of the specified rates. The input data ports shall be MUXed with frame synchronization and necessary overhead information. The composite output data shall be provided, with an accompanying clock to the encoder or binary phase shift key (BPSK) modulator. The composite data and clock shall be provided as a test point via a rear panel output.

3.1.2.4.2 DEMUX. The DEMUX shall accept the demodulated serial data stream from the demodulator or decoder, detect the incoming frame synchronization bits, automatically synchronize itself and reconstruct the three data channels. Each DEMUXed data channel shall be provided to the user interface with an accompanying clock. Two of the output channels shall be



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75 bps TTY data with the third operating at 75 bps, 32 kbps, or 64 kbps selectable by internal switches. The output data on all channels shall be maintained at a continuous mark level when the DEMUX is out of synchronization.

3.1.2.4.3 Output clock stability upon IF dropout or removal. In the out-of-lock condition the stability of the output clock over a period of 2 hours shall be within $\pm 1 \times 10^{-4}$ with a 90 percent confidence.

3.1.3 Functional configuration. The operational configuration of the modem is different at the shore and ship installation, largely due to the data rates in each direction and to the shore-to-ship TDM/broadcast format. The ship or shore interface configuration is shown in FIGURE 1. Without the optional MUX and DEMUX the baseband data interface would simply be a data and clock port with the composite rate of 75 bps to 100 kbps. (The performance range of this model is limited to a maximum symbol rate of 65 kilosymbols per second (ksps) due to modulator bandlimiting). The functional configuration of the modem is shown in FIGURE 2, which illustrates a typical block diagram of a BPSK modem.

3.1.4 External input interface requirements. External input interface requirements shall be as specified in 3.1.4.1 through 3.1.4.2.

3.1.4.1 Demodulator 70-MHz input. The demodulator 70-MHz input shall be as specified in 3.1.4.1.1 and 3.1.4.1.2.

3.1.4.1.1 Input signal characteristics. The 70-MHz demodulator interface signal characteristics shall be consistent with the requirements of 3.2.2.

3.1.4.1.2 Out-of-band interferer signal levels. Modem performance shall not be degraded by out-of-band interferer signals conforming to the signal level characteristics specified in a and b at the demodulator input:

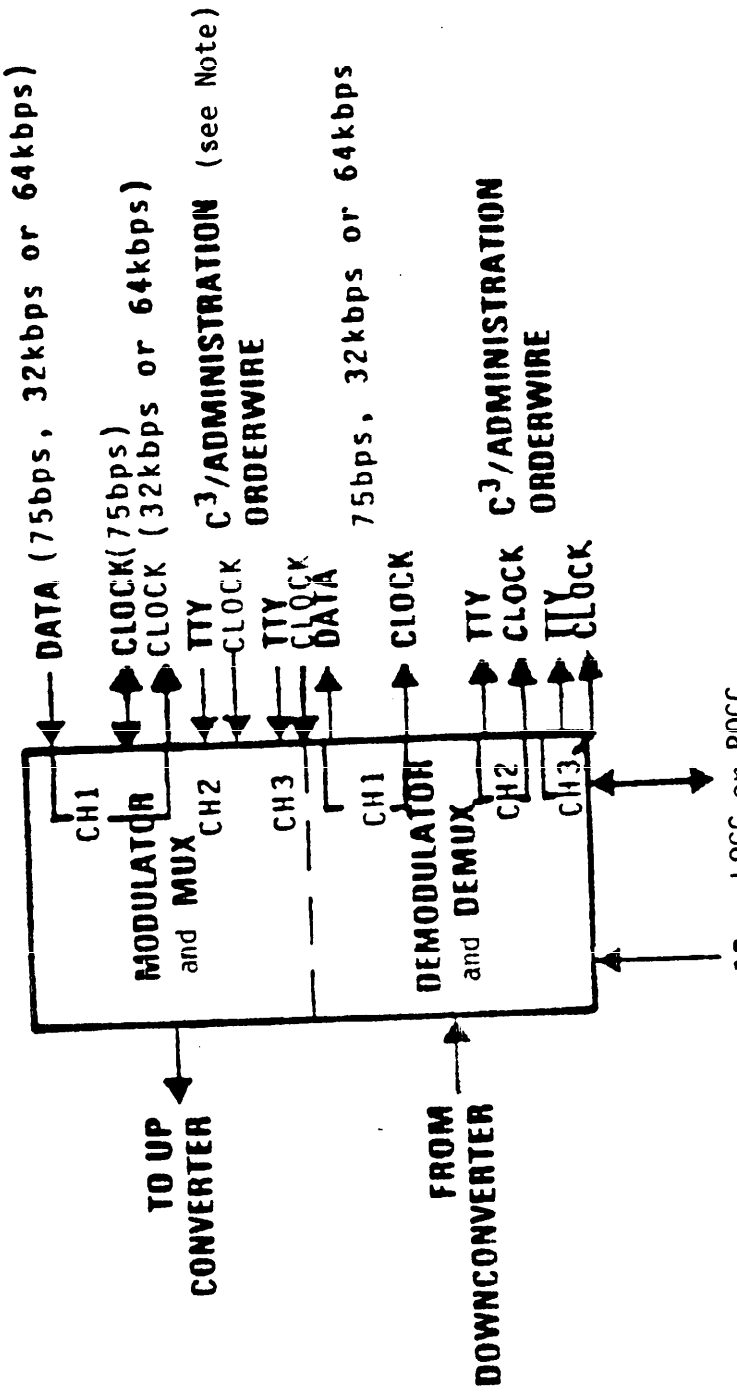
a. Out-of-band interferer spectral signal level components may not exceed a level defined by a slope 20 decibels (dB) below the received signal at 100 kilohertz (kHz) from the received channel center frequency and reducing linearly to 45 dB below the received signal at the channel center frequency.

b. Out-of-band interferer spectral signal level components may not exceed levels defined by slopes increasing linearly at 30 dB/MHz, away from the received signal, in the regions of $f_0$ ±300 kHz to $f_0$ ±1 MHz.

FIGURE 3 illustrates the requirements specified in a and b.



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NOTE: Command, Control and Communications

FIGURE 1. Modem configuration.



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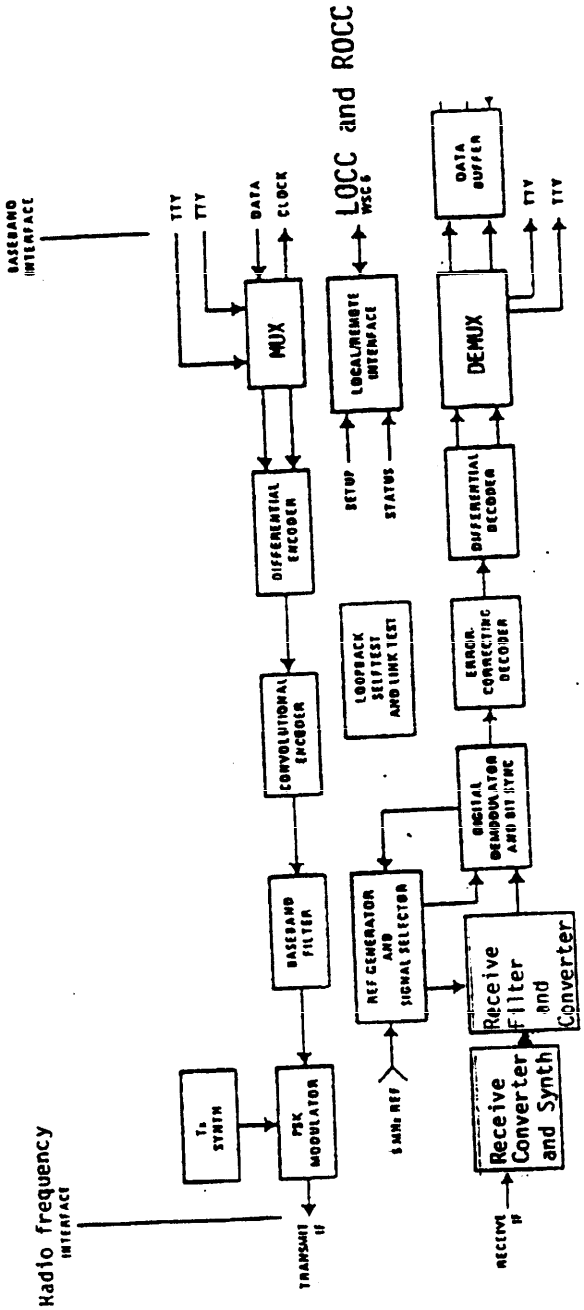



FIGURE 2. Modem simplified functional block diagram.



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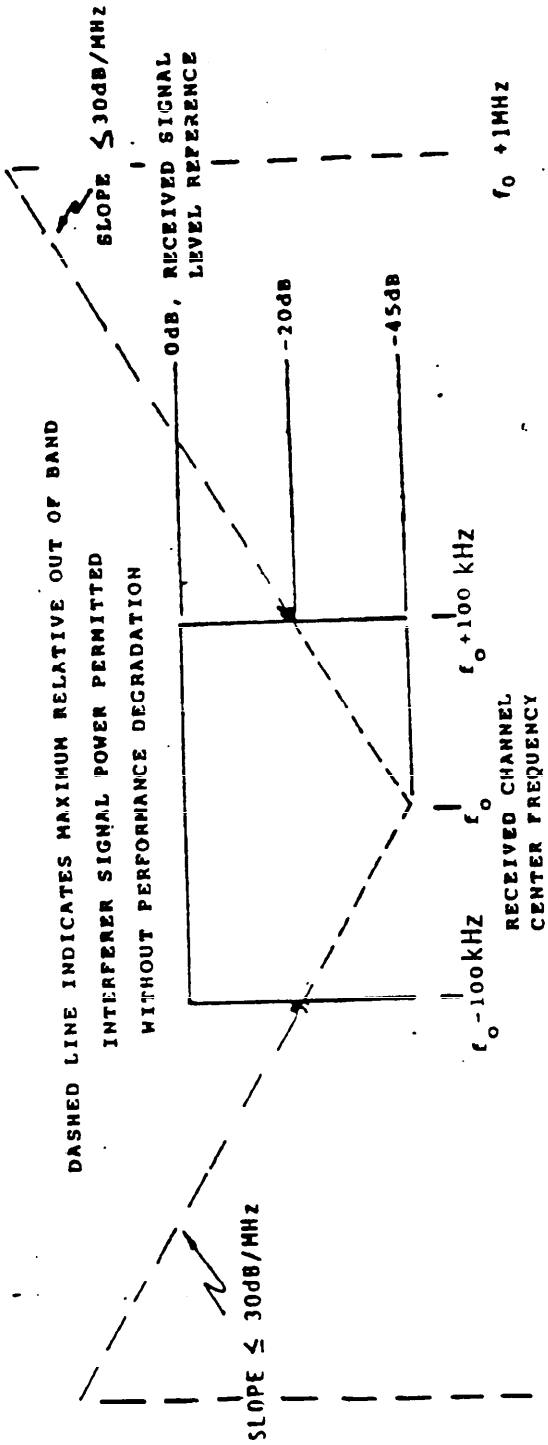



FIGURE 3. Out-of-band interferer signal levels.



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3.1.4.2 5-MHz station reference. A 5-MHz station frequency reference shall be provided to the modem as a standard. This input reference shall conform to or exceed the requirements specified in a through i:

a. Frequency: 5 MHz
b. Stability: $\pm 1 \times 10^{-9}$
c. Level: 0 decibels referred to one milliwatt (dBm) to +13 dBm
d. Impedance: 50 ohms
e. Voltage standing wave ratio (VSWR): 1.3:1
f. Spurious: 40 dB below reference
g. Harmonic distortion: 40 dB below reference
h. Phase noise: >60 dB signal-to-noise ratio (SNR)
i. Connector: TNC

3.2 Performance. Modem performance shall be as specified in 3.2.1 through 3.2.5.6.

3.2.1 Modulator. The modulator shall be as specified in 3.2.1.1 through 3.2.1.11.

3.2.1.1 Digital data input interface. The modem shall accept nonreturn to zero (NRZ) format data with the characteristics specified in a and b from the digital user via a balanced or unbalanced line compatible with MIL-STD-188-114. The modulator shall provide the capability to independently invert the input data or clock, or both, by internal switches. The modem shall be capable of operating in either of two internally-selectable input modes (MUX Bypass Mode only):

a. Mode 1: Operation with a data input or data and clock input, where the modem synchronizes to the data or to the clock if a clock is provided
b. Mode 2: Operation with the modem used as a master clock source

Input mode selection, the data polarity, and clock polarity shall be controlled by internal printed circuit board (PCB) mounted switches.

3.2.1.1.1 Mode 1 operation. In Mode 1, the modem shall accept data from the digital user with a transition density of 12.5 percent to 100 percent and up to 64 consecutive bits without transitions when preceded by 25 percent transition density, and internally regenerate a synchronous clock signal and twice-rate clock signal required for subsequent processing. The input data requirements shall be compatible with MIL-STD-188-114. If an accompanying synchronous clock is provided by the digital user, the modem shall synchronize to the user clock.



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3.2.1.1.2 Mode 2 operation. In Mode 2, the modem shall provide a data rate clock to the digital user, and accept data synchronous with the clock from the digital user. The internally generated clock and twice-rate clock shall be used for subsequent processing. The input data requirements are identical to the Mode 1 requirements. The output clock shall be compatible with MIL-STD-188-114. The modem shall provide the capability to internally invert the data rate clock timing for operation with equipments or in installations where the inverted clock phase is required.

3.2.1.2 Input data rate range. The input data rate range shall be:

a. 75.0 bps to 100 kbps without correction, and
b. 75.0 bps to 5 kbps with error-correcting coding. (Selection of any 3-digit rate within the specified range shall be accommodated.)

The modem shall accept data having an accuracy of $\pm 1 \times 10^{-4}$ for any data rate. (Performance is limited to 65 ksps maximum on this model due to band limiting requirements.)

3.2.1.3 Output frequency. The modem output frequency shall be 69.0 MHz to 71.0 MHz with 100 selections at 20-kHz spacing (to provide 15 frequency division multiplex (FDM) channels at 120-kHz spacing) derived from an externally supplied 5-MHz reference standard. Channelization resolution of 20 kHz shall be provided by front panel control. Accuracy and stability of output frequency shall be consistent with that of the reference standard input.

3.2.1.4 Output level. The modem output level shall be 0 dBm +6 dB, -3 dB.

3.2.1.5 Output impedance. The output impedance shall be 50 ohms nominal with less than 1.5:1 VSWR over the output frequency range of 70 MHz ±0.9 MHz.

3.2.1.6 Output modulation. Output modulation shall be biphase (0 degree, 180 degree) PSK. A zero-data bit shall generate a 0 degree carrier phase and a one data bit shall generate a 180 degree switch in the carrier phase.

3.2.1.7 Antipodal keying error. Antipodal keying error shall be ±5 degrees maximum.

3.2.1.8 Amplitude modulation (AM) of signal component. Incidental AM of the signal component shall be 0.2 dB maximum.

3.2.1.9 Output spectrum. All output harmonic and spurious responses shall be 40 dB or more below the unmodulated carrier. The modulated signal sideband levels shall be $\geq$24 dB below the unmodulated carrier level at an offset of ±100 kHz from the channel center frequency to provide adjacent channel interference protection.



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3.2.1.10 Encoding. Both internal differential (NRZ-M or NRZ-S) coding and internal one-half rate convolutional coding shall be available. Coders may be individually bypassed by internal controls. Status indication shall be displayed on the front panel.

3.2.1.10.1 Encoding switch. Differential encoding shall be a switch-selectable option which shall always be in effect when MUXing functionality is employed. The modem shall include a switch to select differential MARK encoding (NRZ-M) and differential SPACE encoding (NRZ-S).

3.2.1.10.2 One-half rate evolutional coding. One-half rate evolutional coding shall be as shown in FIGURE 4.

3.2.1.11 Output clock. The output data rate clock shall be in accordance with MIL-STD-188-114, balanced or unbalanced compatible. The frequency shall be the programmed data rate to an accuracy within $\pm 1 \times 10^{-8}$.

3.2.2 Demodulator. The demodulator shall be as specified in 3.2.2.1 through 3.2.2.25.3.

3.2.2.1 Input frequency. The input frequency shall be 69.0 MHz to 71.0 MHz with 100 selections at 20-kHz spacing (to provide 15 FDM channels at 120-kHz spacing) derived from an externally supplied 5-MHz reference standard. Channelization resolution of 20 kHz shall be provided by front panel control.

3.2.2.2 Input impedance. The input impedance shall be 50 ohms nominal with less than 1.5:1 VSWR over the input frequency range of 70 MHz ±0.9 MHz.

3.2.2.3 Input modulation. Input modulation shall be biphase (0 degree, 180 degree) PSK. A zero data bit shall generate a 0 degree carrier phase and a one data bit shall generate a 180 degree switch in the carrier phase.

3.2.2.4 Antipodal keying error. Antipodal keying error shall be ±5 degrees maximum.

3.2.2.5 AM of signal component. Incidental AM of the signal component shall be 0.2 dBm maximum.

3.2.2.6 Input noise spectral density. The modem design shall accommodate an input spectral density of -95 dBm/Hz ±10 dB.

3.2.2.7 Minimum input signal power. The receiver shall be capable of demodulating a signal level of -76 dBm, corresponding to an $E_S/N_O$ of 4.0 dB at an input symbol rate of 300 bps or -80 dBm corresponding to an $E_S/N_O$ of 6.0 dB at an input bit rate of 75 bps.



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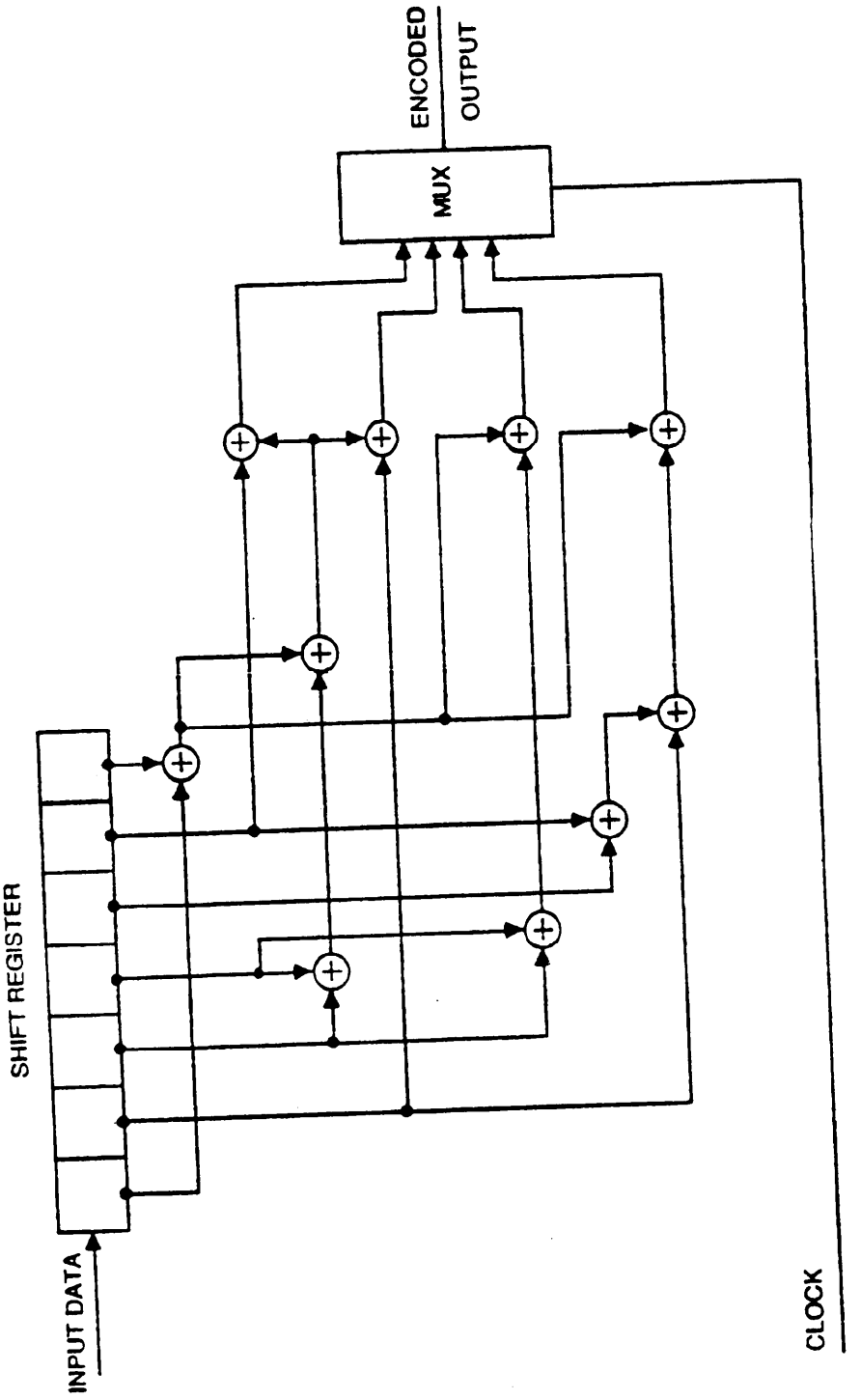



FIGURE 4. Functional block diagram of a convolutional encoder.



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3.2.2.8 Maximum total IF input power. The maximum total IF input power shall be -10 dBm (signal-plus-noise).

3.2.2.9 Nondamage input level. The nondamage input level shall be +20 dBm.

3.2.2.10 Image rejection. Image rejection shall be 36 dB minimum.

3.2.2.11 Signal acquisition range. The signal acquisition range shall be ±6 kHz maximum from selected IF (selectable ranges as specified in TABLE I).

TABLE I. Operating controls and indicators.

| Function | Description |
|---|---|
| Differential encoder Mode Select | Internal switches; selects operation with NRZ-M differential encoding, NRZ-S differential encoding, or no differential encoding. The selected mode shall be indicated on front panel indicators. |
| Differential decoder Mode Select | Internal switches; selects operation with NRZ-M differential decoding, NRZ-S differential decoding, or no differential decoding. The selected mode shall be indicated on front panel indicators. |
| Convolutional encoder Mode Select | Internal switch; selects encoder operation with one-half rate transparent error-correction code or no error correction encoding. The selected mode shall be indicated on front panel indicators. |
| Error-correcting decoder Mode Select | Internal switch; selects decoder operation with one-half rate transparent code, or no error correction decoding. The selected mode shall be indicated on front panel indicators. |



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TABLE I. Operating controls and indicators - Continued.

| Function | Description |
|---|---|
| Transmit data rate Select | Front panel switches; provides selection of the operating rate of the modulator functions over the input data rate range of 75 bps to 100 kbps (75 bps to 50.0 kbps if error-correcting encoding is selected). |
| Receive data rate Select | Front panel switches; provides selection of the operating rate of the demodulator and bit sync functions over the input data rate range of 75 bps to 100 kbps (75 bps to 50 kbps if error-correcting decoding is selected). |
| Carrier offset control | Front panel switches; controls the operating center frequency of the demodulator carrier recovery loop. Polarity and two-digit magnitude selection is provided in 10-Hz steps from +490 Hz to -490 Hz referred to the demodulator input center frequency selected. |
| Sweep limit control | Front panel switch; controls the acquisition range of the demodulator. Provides operation with the sweep ranges referred to the IF center frequency of:<br>±6 kHz<br>±3 kHz<br>±1.5 kHz<br>±750 Hz<br>±375 Hz |
| Loop offset indicator | Front panel indicator; provides an indication of the uncompensated offset existing in the demodulator carrier recovery loop when the input signal has been acquired. |



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TABLE I. Operating controls and indicators - Continued.

| Function | Description |
|---|---|
| Carrier lock indicator | Front panel indicator; illuminates when demodulator carrier recovery loop has phase-locked to the input 70-MHz IF signal. |
| AC power control | Front panel switch; controls the AC power input to the unit. |
| AC power indicator | Front panel indicator; illuminates when AC power is ON. |
| Carrier lock status relay | Rear panel connector provides a relay closure when the demodulator carrier recovery loop has phase-locked to the input IF signal. The relay shall be capable of controlling 24 V at 0.25 amperes. |
| Interface configuration controls | Internal switches; controls digital user interface signal configuration at initial installation; consists of internal PCB-mounted switches as follows:<br><br>Input data polarity - selects normal or inverted polarity of modulator input data.<br><br>Input clock polarity - selects normal or inverted polarity of modulator input clock or internally generated clock.<br><br>Output data polarity - selects normal or inverted polarity of demodulator output data.<br><br>Output clock polarity - selects normal or inverted polarity of demodulator output clock. |



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TABLE I. Operating controls and indicators - Continued.

| Function | Description |
|---|---|
| | Transmit Bit Sync Mode Select - selects Mode 1 or Mode 2 operating configuration as specified in 3.1.2.1. |
| Relative signal quality indicator | A digital display of relative signal quality which can be correlated to the margin with respect to an acceptable BER. |
| Transmit channel select | Front panel switches; provides selection of modulator output channel with 20 kHz resolution (100 channels, 20 kHz each, 0 through 99). |
| Receive channel select | Front panel switches; provides selection of demodulator input channel (100 channels 20 kHz each, 0 through 99). |
| Lamp test | Front panel pushbutton switch to test all lamps. This shall not affect any operational modes. |
| Loop bandwidth select | Internal switch; provides selection of carrier recovery loop bandwidth for operation at narrow or wide loop bandwidth. The selected mode shall be indicated on front panel indicators. |
| Local/remote control | Front panel switch; provides selection of operating functions as specified in 3.2.3.2 from front panel in Local Mode or terminal control console in Remote Mode. The Remote selection shall be evidenced by a front panel indicator. |
| Transmit input bit synchronizer lock indicator | Front panel indicator; provides an indication that the demodulator bit synchronizer is locked to the received data when illuminated. |



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TABLE I. Operating controls and indicators - Continued.

| Function | Description |
|---|---|
| Receive demod bit sync lock indicator | Front panel indicator; provides an indication that the demodulator bit synchronizer is locked to the received data when illuminated. |
| Agile IF synthesizers, transmit and receive, lock indicator | Two front panel indicators; provides an indication that the agile IF synthesizers are locked to the selected frequency when illuminated. |
| Summary fault | Front panel indicator; illuminates when any of these functions are in a fault condition:<br>Transmit IF synthesizer lock<br>Receive IF synthesizer lock<br>MUX synthesizer lock<br>DEMUX synthesizer lock<br>Overtemperature<br>DC power |
| Summary fault relay | Rear panel connector provides a relay closure when a summary fault condition exists. The relay shall be capable of controlling 24 V at 0.25 amperes. |
| Audible alarm | Front panel alarm; emits audible signal when the carrier recovery loop is in an out-of-lock condition. |
| MUX/DEMUX controls and indicators | |
| MUX select | Front panel switch; selects operation with MUX or Bypass Mode. |
| MUX format | Internal switch; selects MUX Format Mode 75-bps, 32-bps, or 64-kbps format. The format selected shall be indicated on the front panel. |
| DEMUX select | Front panel switch; selects operation with DEMUX or Bypass Mode. |



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TABLE I. Operating controls and indicators - Continued.

| Function | Description |
|---|---|
| DEMUX format | Internal switch; selects DEMUX Format Mode, 3-, 7-, or 17-channel input with channel 1 at 75 bps, 32 bps, or 64 kbps. The format selected shall be indicated on the front panel. |
| TDM 75 bps data channel select | Internal switch; provides DEMUX 75 bps data channel selection (1 of 17 channels). |
| TDM $C^2$ TTY channel select | Internal switch; provides DEMUX OW/TTY 1/ channel selection (1 of 17 channels). |
| TDM OW/TTY channel select | Internal switch; provides DEMUX OW/TTY channel selection (1 of 17 channels). |
| MUX synthesizer lock indicator | Front panel indicator; provides an indication that the MUX synthesizer is locked when illuminated. |
| DEMUX synthesizer lock indicator | Front panel indicator; provides an indication that the DEMUX synthesizer is locked when illuminated. |
| DEMUX group sync lock indicator | Front panel indicator; provides an indication that the DEMUX is locked to the received data when illuminated. |
| DEMUX group sync lock status relay | Rear panel connector provides a relay closure when the DEMUX is locked to the received data. The relay shall be capable of controlling 24 V at 0.25 amperes. |

1/ Orderwire teletype



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3.2.2.12 Signal acquisition time ($T_a$). The modem shall acquire the input signal, with a 90 percent confidence level within a time $T_a$ after application of the input. The $T_a$ for the selectable frequency ranges specified in TABLE I at an input symbol rate of 600 symbols per second under the specified conditions shall not exceed the times specified in TABLE II.

TABLE II. $T_a$ (seconds).

| Input frequency | Loop bandwidth | |
|---|---|---|
| Range (kHz) | Narrow | Wide |
| ±6.0 | 566 | 98 |
| ±3.0 | 283 | 49 |
| ±1.5 | 142 | 24.5 |
| ±0.750 | 71 | 12.3 |
| ±0.375 | 35.5 | 6.1 |

3.2.2.13 Carrier lock threshold. The mean-time-to-loss of demodulator lock after signal acquisition shall be greater than $10^7$ symbols at any $E_S/N_O$ above 4.0 with narrow loop bandwidth and without Doppler acceleration or adjacent channel interference applied for symbol rates >300 bps. For symbol rates below 300 bps, the required $E_S/N_O$ to maintain the threshold performance shall increase proportional to symbol rate reduction to 6.0 dB at 75 bps. (See 3.2.3.25 for carrier lock threshold performance with applied distortion.)

3.2.2.14 Bit timing loop acquisition. Bit timing loop acquisition shall be as specified in 3.2.2.14.1 and 3.2.2.14.2.

3.2.2.14.1 Acquisition range. Acquisition range shall be ±0.025 percent of the programmed data rate.

3.2.2.14.2 Acquisition time. Loop phase error shall be resolved within 10 degrees in less than 10,000 bit periods (random data).

3.2.2.15 Bit timing loop input transition density. The bit timing loop input transition density shall be from 12.5 percent to 100 percent and up to 64 consecutive bits without transitions when preceded by 25 percent transition density.

3.2.2.16 Bit timing loop input isochronous distortion. The modem shall accept data with 20 percent maximum isochronous distortion.

3.2.2.17 Bit count integrity. The probability of adding or losing an information bit on a per bit basis shall be $<10^{-12}$.



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3.2.2.18 Output data. Output data shall be NRZ format, MIL-STD-188-114 balanced and unbalanced compatible. Channel 1 data shall be provided with synchronous clock to the rear panel interface connector. The Channel 1 data shall be rate buffered and provided to the rear panel interface connector when clocked with an external asynchronous clock with $\pm 2 \times 10^{-9}$ of nominal rate. The data buffering shall be sufficient to provide bit count integrity for $\geq$24 hours at data rates $\leq$64.0 kbps and in the presence of link rate offsets $\pm 1 \times 10^{-7}$.

3.2.2.19 Output clock. The output data rate clock shall be synchronous with data, MIL-STD-188-114 balanced and unbalanced compatible. The duty cycle shall be 50/50 ±10 percent. With normal polarity the positive clock transition shall be coincident with the output data transition ±1.0 microsecond (µs). In the out-of-lock condition the stability of the output clock over a period of 2 hours shall be within $\pm 1 \times 10^{-4}$ with a 90 percent confidence.

3.2.2.20 Root-mean-square (rms) output timing jitter. The rms output timing jitter shall be 4 percent maximum.

3.2.2.21 Output data rate range. The output data rate range shall be 75.0 bps to 50 kbps with error correcting decoding (MUX Bypass Mode) and 75.0 bps to 100 kbps without error correcting decoding (MUX Bypass Mode). Selection of any 3-digit rate within the specified ranges is accommodated.

3.2.2.22 Decoding. Both internal differential (NRZ-M or NRZ-S) decoding and internal one-half rate maximum decoding shall be available. Decoders may be individually bypassed by internal controls. Status indication shall be displayed on the front panel. Differential decoding shall be a switch-selectable option which shall always be in effect when the DEMUXing functionality is employed.

3.2.2.23 Coding gain. The coding gain shall be $\geq$4.6 dB at bit error rate (BER) = $1 \times 10^{-5}$.

3.2.2.24 Demodulator bit synchronizer and decoder performance. The BER performance of the demodulator bit synchronizer without Doppler acceleration or adjacent channel interference applied, with and without an error-correcting decoder, shall be within 1.5 dB of the appropriate curve as shown in FIGURE 5 using the narrow carrier recovery loop bandwidth for symbol rates $\leq$65 ksps. When the wide loop bandwidth is employed, the error correcting coder/decoder performance will be degraded. Performance may be limited to 65 ksps maximum due to bandlimiting modulator requirements.

3.2.2.25 Performance with applied distortion. The modem shall be capable of operating under Doppler stress with peak acceleration of 2.0 gs (within a period of 5 seconds) for symbol rates $\geq$600 bps or adjacent channel interference from two simultaneous interferers spaced ±120 kHz about the received channel.



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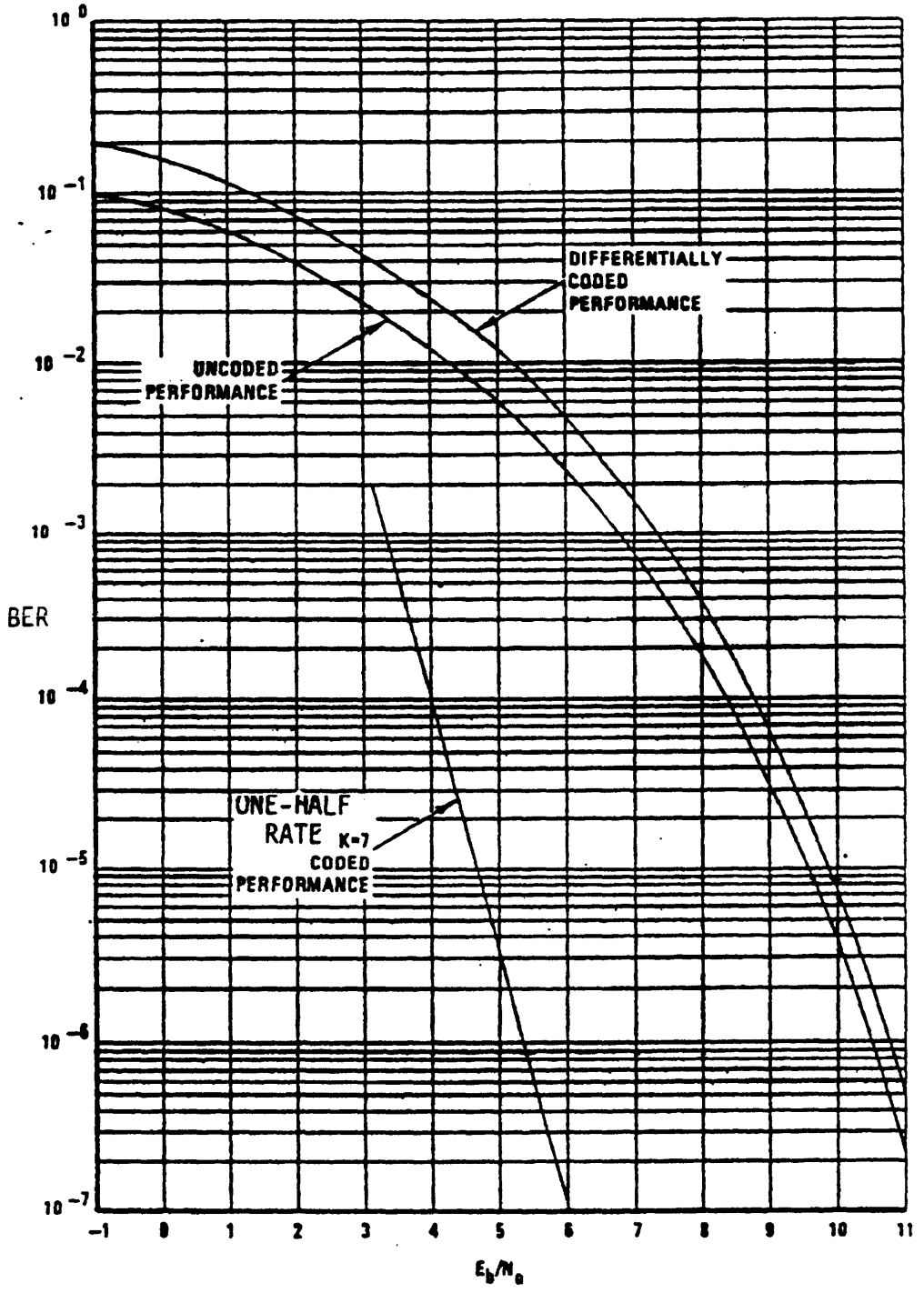



FIGURE 5. BER performance.



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TABLE III. Doppler acceleration performance.

| Symbol rate | Loop bandwidth | Implementation loss (BER relative to theoretical) | Carrier lock threshold ($E_b/N_o$) | Peak Doppler acceleration (5-second period) |
|---|---|---|---|---|
| ≥3.00 kbps | Narrow | 1.5 dB | 4.0 dB | 0.0 g |
| 1.00 kbps to 2.99 kbps | Wide | 1.5 dB | 9.0 dB | 0.0 g |
| 600 bps to 999 bps | Wide | 2.0 dB | 11.0 dB | 0.0 g |
| 300 bps to 599 bps | Wide | 2.0 dB | 11.0 dB | 0.0 g |
| ≥3.00 kbps | Narrow | 2.0 dB | 6.0 dB | 2.0 g |
| 1.00 kbps to 2.99 kbps | Wide | 2.0 dB | 11.0 dB | 2.0 g |
| 600 bps to 999 bps | Wide | 2.5 dB | 13.0 dB | 2.0 g |
| 300 bps to 599 bps | Wide | 2.5 dB | 15.0 dB | 0.5 g |





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3.2.2.25.1 Doppler acceleration performance. The capability to select various loop bandwidths by an internal switch shall be provided to minimize performance degradation due to Doppler effects. Front panel indicators shall be provided to indicate selected mode. The performance shall be within the limits specified in TABLE III.

3.2.2.25.2 Adjacent channel interference performance. With two adjacent channel interferer signals up to 6 dB greater in level than the received signal and sourced by MD-1030A(V) type modulators, the performance shall be within the limits specified in TABLE IV. For interferer symbol rates between 65 ksps and 100 ksps, the minimum channel spacing to maintain specified performance shall be ±1.85 x symbol rate.

TABLE IV. Interferer signal characteristics.

| Symbol rate | Minimum spacing about received signal | Loop bandwidth | Implementation loss (BER relative to theoretical) | Carrier lock threshold ($E_b/N_o$) |
|---|---|---|---|---|
| ≤65 ksps | ±120 kHz | Narrow | 3.0 dB | 5.5 dB |
| ≤65 ksps | ±120 kHz | Wide | 3.5 dB | 10.5 dB |

3.2.2.25.3 Data buffering. The modem shall be equipped with an internal buffer to compensate for Doppler-induced frequency offsets. The buffering shall be sufficient to prevent clock slips over a 24-hour period.

3.2.3 Local operation and control center (LOCC) and remote operation and control center (ROCC) interface. The modem shall provide a microprocessor-controlled, remote control and status, bus interface with the AN/WSC-6(V) terminal control console. The interface circuitry shall be implemented as specified in NRL Drawing (DWG) 7259302-112.

3.2.3.1 Remote status. The modem front panel status specified in a through l and in TABLE I and TABLE V shall be provided at all times to the terminal control console:

a. Local/Remote Mode switch selection
b. Bandwidth switch selection
c. Sweep limit control switch selection
d. Carrier lock indicator status
e. Agile IF channel selection (transmit and receive)
f. One-half rate encoder (ON or OFF) status
g. One-half rate decoder (ON or OFF) status
h. Differential encoder (NRZ-M, NRZ-S, or OFF) status
i. Differential decoder (NRZ-M, NRZ-S, or OFF) status
j. Fault summary signal
k. Relative signal quality status
l. DEMUX status outputs shall be provided as TDM DEMUX lock status



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3.2.3.2 Remote control. The modem functions specified in a through d shall be under direct bus control by the AN/WSC-6(V) control console when in the Remote Mode:

a. Bandwidth selection
b. Sweep limit selection
c. Agile IF channel selection (transmit and receive)
d. Transmit data rate and receive symbol rate selection

(excluding self-test). Remote control selections will not be displayed on the modem front panel.

TABLE V. Test controls.

| Function | Description |
|---|---|
| Unit operating mode control | Front panel switch; configures the unit either for normal or test operation.<br><br>Operate position - configures the unit for normal operation.<br><br>Test position - configures the unit for test operation. |
| Test control | Front panel switch; enabled only when the unit is in Test Mode - configures the unit either for loop-back or link testing.<br><br>Loop position - energizes relay to connect modulator output to demodulator input (via an attenuator) and to interrupt normal 70-MHz IF interfaces.<br><br>Link position - provides normal 70-MHz IF interfaces. |



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TABLE V. Test controls - Continued.

| Function | Description |
|---|---|
| | The internal test generator and error comparator shall provide a data pattern compatible with the 511 ($2^9$-1) sequence generated by utilizing linear feedback shift register tap 5 and tap 9 to provide compatibility with the HP-1645A BER test set. |
| Sync monitor | Front panel signal interface; enabled only when the unit is in Test Mode. Buffered TTL[1/] compatible pulse occurring each time the internally generated pseudo-random sequence repeats (sequence length 511 bits). Pulse width equal to period of modulator input data rate control switch setting. |
| Clock monitor | Front panel signal interface; buffered TTL compatible output corresponding to the demodulator output clock |
| Error monitor | Front panel signal interface; buffered TTL compatible output. When the unit is in the Test Mode, this output consists of a pulse occurring each time an output error is detected by the error comparator. |
| Test loop | Front panel switch; selects data and clock inputs to the error detector from various modem functional circuit groups. |
| DC power supply indicator | Front panel indicator. Illuminated when DC power is within limits. |



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TABLE V. Test controls - Continued.

| Function | Description |
|---|---|
| GO indicator | Front panel indicator illuminates to indicate a GO or PASS condition of a test result when the unit is in the Test Mode. |
| Reset | Front panel push button switch. When the unit is in the Test Mode, the GO indicator latch is reset for a test result indication when the switch is momentarily depressed. |
| Single error | Front panel push button switch inserts a single error into the PN test sequence when depressed. |

1/ Transistor-transistor logic

3.2.4 MUX. The modem shall provide a MUX which simultaneously accepts asynchronous or isochronous serial data from two TTY inputs and synchronous data from one computer input with or without an accompanying clock. The MUX shall buffer the input data, insert frame synchronization and clock phasing information, and MUX these data into a serial data stream for transmission. The MUX shall be capable of supplying a clock to the user for the synchronous data channel only.

3.2.4.1 Input data rate. Input data rates shall be two each asynchronous or isochronous data channels at 75 bps TTY and one synchronous data channel, internal switch-selectable for 75 bps, 32 kbps, or 64 kbps.

3.2.4.2 Input data rate accuracy. Input data rate accuracy shall be at least ±0.01 percent ($\pm 10^{-4}$).



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3.2.4.3 Data clock interface. The data clock interface shall handle data with NRZ format. The clock shall be 50/50 ±10 percent duty cycle, with a positive clock transition coincident with data transition ±1.0 µs. Jitter on the input data and clock shall not exceed 5 percent of the nominal data unit interval. The data clock shall be compatible with MIL-STD-188-114, balanced or unbalanced.

3.2.4.4 Composite transmit rates. Composite transmit rates shall be as specified in TABLE VI.

TABLE VI. Composite transmit rates.

| Selected rate | Overhead (maximum) |
|---|---|
| 600 bps | 62.5 percent |
| 32.3 kbps | 2 percent |
| 64.6 kbps | 2 percent |

3.2.5 DEMUX. The modem shall provide a DEMUX which accepts a TDM serial data stream from the demodulator, synchronizes to this data stream and selects the appropriate three channels (internal switch-selectable) which pertain to a specific link while ignoring additional channels.

3.2.5.1 Modes/output rates. The modem shall handle the five DEMUXing modes specified in TABLE VII.

TABLE VII. DEMUXing modes.

| Mode of operation | | Composite rate | Channel 1 rate | Other channels rate | |
|---|---|---|---|---|---|
| Mode | Function | | | Quantity | |
| 1 | Three-channel | 600 bps | 75 bps | 2 | 75 bps |
| 2 | Seven-channel | 600 bps | 75 bps | 6 | 75 bps |
| 3 | 17-channel | 1350 bps | 75 bps | 16 | 75 bps |
| 4 | 32k data rate | 32.3 kbps | 32 kbps | 2 | 75 bps |
| 5 | 64k data rate | 64.6 kbps | 64 kbps | 2 | 75 bps |



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3.2.5.1.1 Seventeen-Channel Mode. The Seventeen-Channel Mode is shown in FIGURE 6. Each frame begins with 24 bits of sync information followed by 24 bits from each of the 17 channels. Since each channel is nominally 75 bps and there are effectively 18 channels (including the sync word), the overall transport rate is 18 times 75 bps or 1350 bps. Since all channels are asynchronous to one another and to the transport rate, they may vary in data rate by ±0.01 percent, necessitating the use of stuff bits to compensate for rate discrepancies.

3.2.5.1.1.1 Channel input rate acceleration. When a channel input is running slightly slower than 75 bps with respect to the transport rate, it is necessary occasionally to omit a bit to allow the input to catch up with the transport rate. This minus stuff operation is accomplished by inserting only 23 valid data bits during the channel allocated time period. A twenty-fourth dummy bit is always transmitted but is later discarded by the receiving DEMUX. When a channel is running slightly faster than 75 bps with respect to the transport rate, it is necessary occasionally to transmit a twenty-fifth bit. This plus stuff operation inserts a twenty-fifth bit into the twenty-fourth bit location of the sync information. The receiving DEMUX later retrieves this extra bit and inserts it into the appropriate channel data stream.

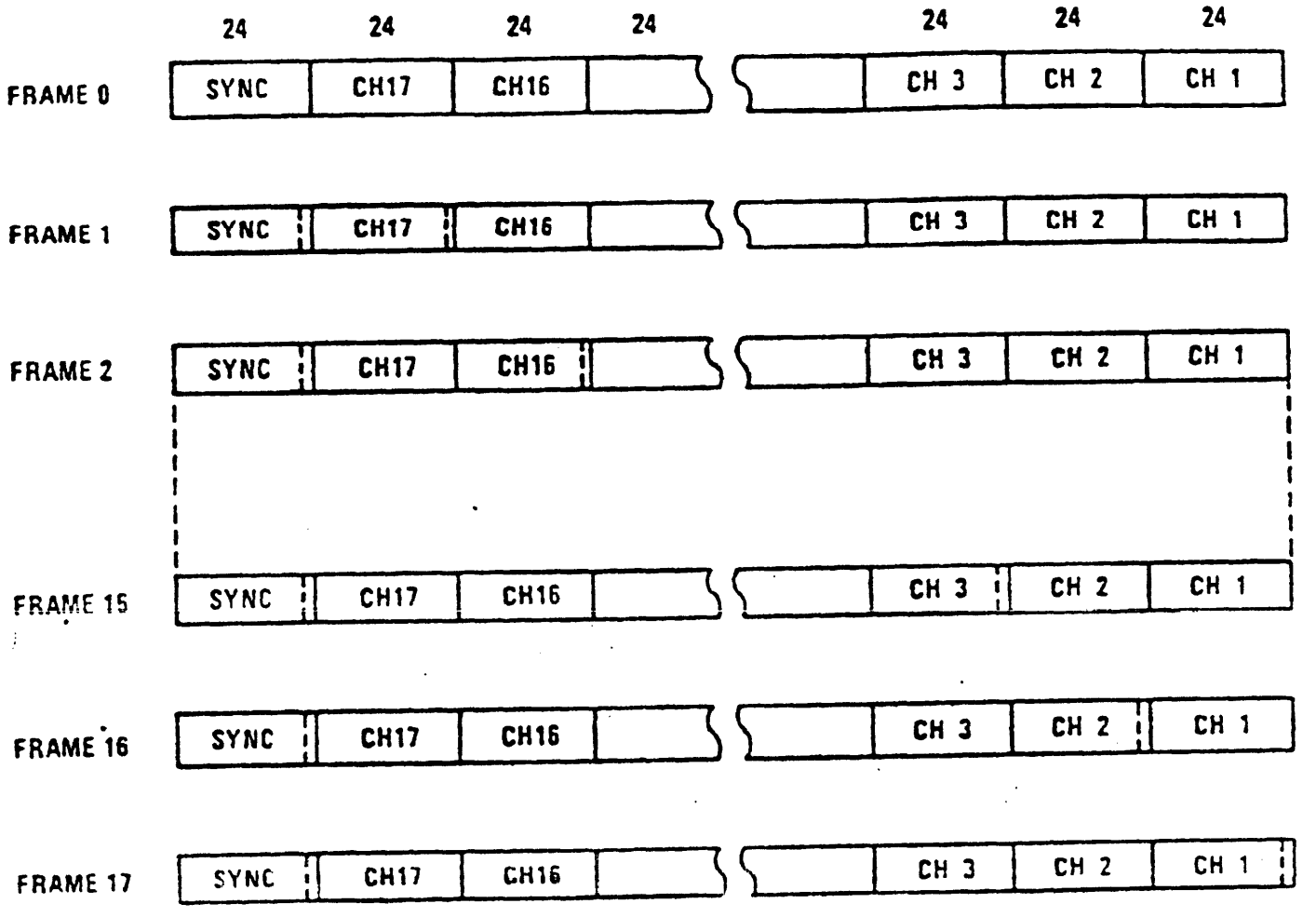
3.2.5.1.1.2 Stuff-bit operation. Each frame allows one channel an opportunity for a stuff bit operation. Each consecutive frame allows a stuff bit to occur for the next channel such that over a period of 17 frames all 17 channels have had the opportunity for stuff bit servicing. The beginning of this superframe is denoted by the occurrence of a special frame in which no stuff bit operations are allowed.

3.2.5.1.2 Seven-Channel Mode. The Seven-Channel Mode shown in FIGURE 7 is very similar to the Seventeen-Channel Mode. The same 24-bit blocks of data are used to MUX seven channels of data to create a frame which is only 192 bits in length and requires a transport rate of eight times 75 bps or only 600 bps. Following the superframe designator (Frame 0), only seven frames are required to supply stuff bit servicing for all channels. As a result, the superframe is only eight frames or 1536 bits in length.

3.2.5.1.3 Three-Channel Mode. The Three-Channel Mode shown in FIGURE 8 is identical to the seven-channel format with one exception. Four available channel allocations are not used. Dummy data patterns are inserted on what would be Channel 4 and Channel 6 to increase the transition density of the composite data stream. Since Channel 4 through Channel 7 are not used, no stuff bit servicing is performed in Frame 1 through Frame 4. Frame 5 and Frame 6 allow stuff bit servicing of Channel 3 and Channel 2, respectively. Channel 1 is the modem main data channel and requires no stuff bit servicing since it is always synchronous to the modem transport rate. When operating with this format, the modem transmit data rate selector should be set to the composite data rate of 600 bps even though the inputs are only 75 bps.



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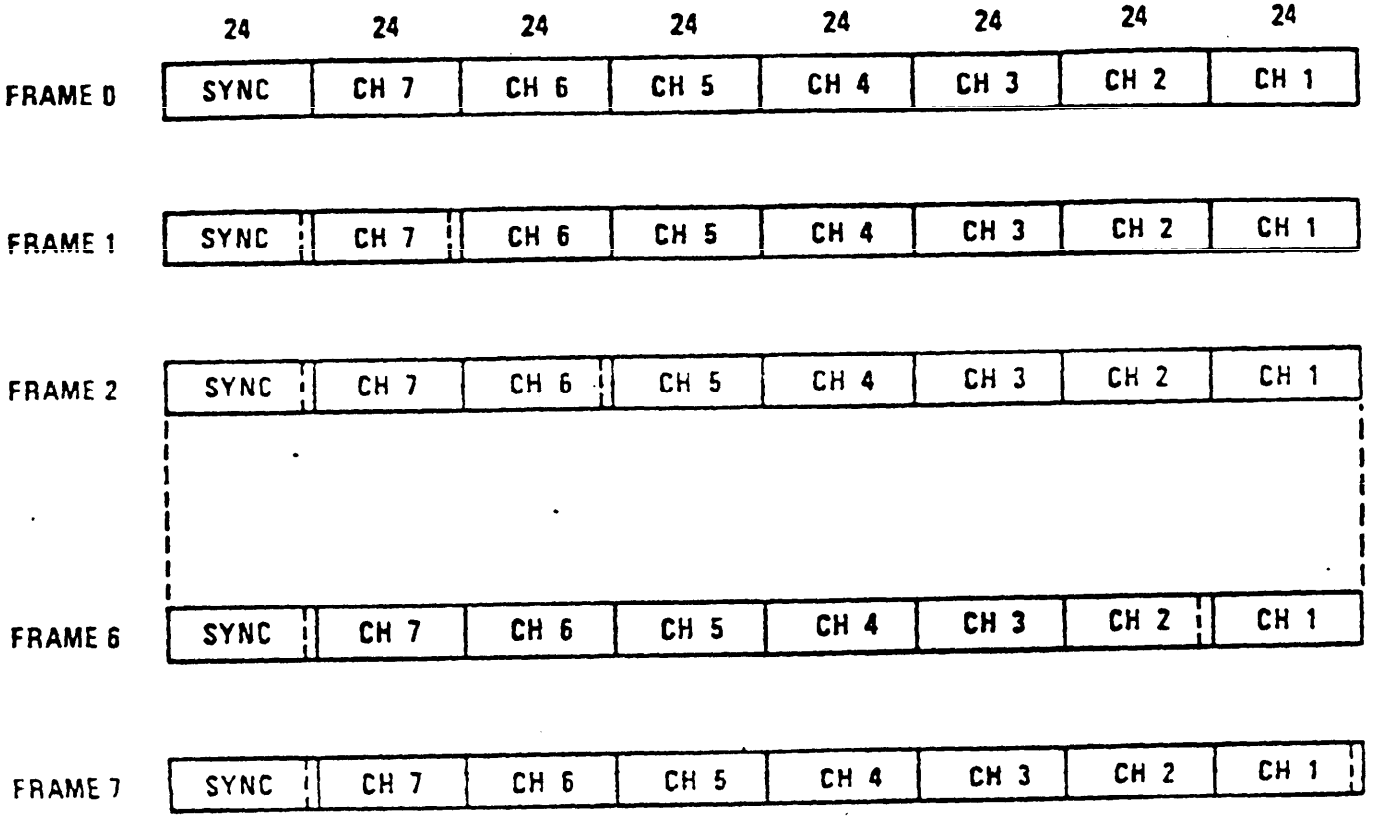



Frame length - 432 Bits
Superframe length - 7776 Bits
Transport rate - 1350 bps

FIGURE 6. Seventeen-Channel Mode.



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Frame length - 192 Bits

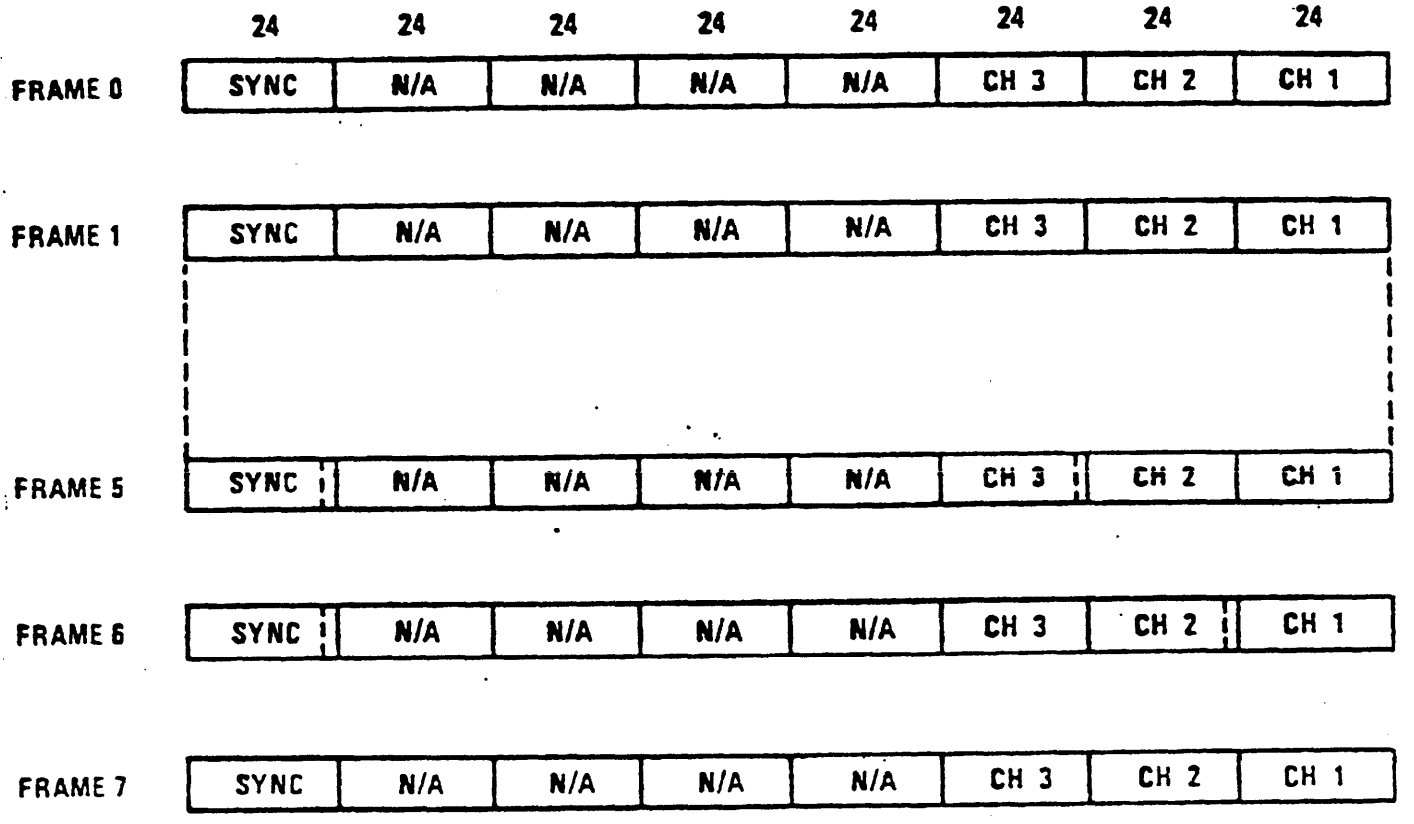
Superframe length - 1536 Bits

Transport rate - 600 bps

FIGURE 7. Seven-Channel Mode.



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Frame length - 192 Bits

Superframe length - 1536 Bits

Transport rate - 600 bps

FIGURE 8. Three-Channel Mode.



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3.2.5.1.4 32 kbps-Channel Mode. The 32 kbps-Channel Mode shown in FIGURE 9 is slightly different from those specified 3.2.5.1.1 through 3.2.5.1.3. The relative number of bits allocated for the higher rate channel and the lower rate channels shall conform to the ratio 75:32000. To maximize similarity to other formats, the same 24 bits of sync information and 24 bits of channel data for the 75-bps channels are used, requiring 10,240 bits for the 32-kbps channel. The total composite data rate required for sync information, two 75-bps channels and one 32 kbps channels, would be 32,000 bps +3 times two 75 bps, or 32,225 bps. Since this is not a convenient rate for the modem to synthesize, a fourth dummy channel of 24 bits per frame is used to increase the composite rate to an even 3-digit data rate of 32.3 kbps. Frame 0 identifies the beginning of the superframe. Frame 1 contains no stuff bit servicing (dummy channel is not used). Frame 2 and Frame 3 contain stuff bit servicing for Channel 3 and Channel 2, respectively. Channel 1 is always synchronous to the modem transport rate and does not require stuffing. In this mode of operation the transmit data rate should be set to 32.3 kbps. A 32.0 kbps transmit reference clock is supplied to the user interface to transfer main channel data from the user to the modem.

3.2.5.1.5 64 kbps-Channel Mode. The 64 kbps Channel Mode shown in FIGURE 10 is similar to the 32 kbps format. In this mode the main channel data rate is doubled while the 75 bps channels remain the same. This is accomplished by doubling the superframe length (repeating the 32 kpbs format) and transport rate while using the 24-bit allocations for Channel 2 and Channel 3 half of the time. Only odd numbered frames contain Channel 3 data with Frame 7 handling stuff bit servicing for that channel. Only even numbered frames contain data from Channel 2 with Frame 8 handling stuffing operations for Channel 2. Channel 4, the dummy channel, appears in every frame and does not require stuff bit servicing. Channel 1 is synchronous to the modem transport rate and requires no stuffing. In this mode, the transmit data rate shall be set to 64.6 kbps and a 64.0-kbps clock is supplied to the user interface as a main channel transmit reference clock.

3.2.5.2 Output data rate accuracy. The output data rate accuracy shall be at least 99.99 percent.

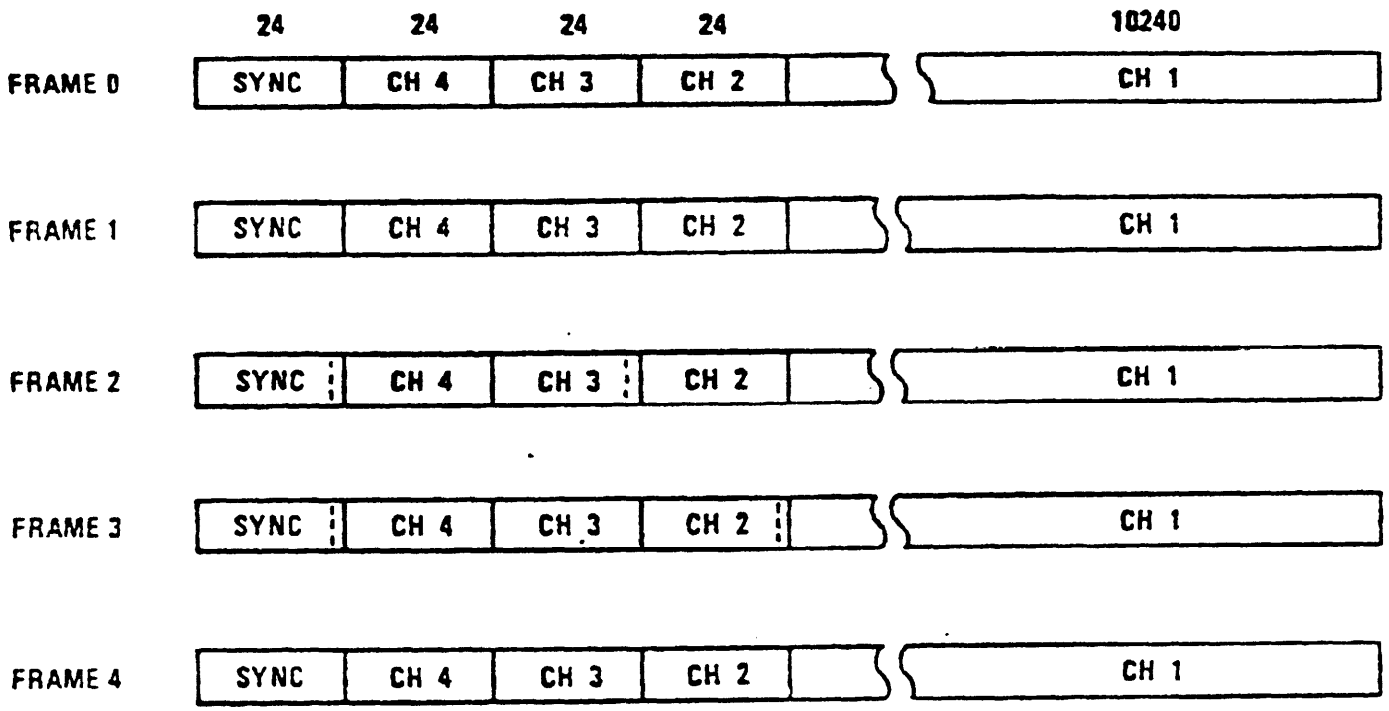
3.2.5.3 Data and clock interface. The data and clock interface shall be NRZ format. The clock shall be 50/50 ±10 percent duty cycle, with a positive clock transition coincident with data transition ±1.0 µs. Jitter on the output data and clock shall not exceed 5 percent of the nominal data unit interval timing. The data and clock outputs shall be compatible with MIL-STD-188-114, balanced or unbalanced.

3.2.5.4 Acquisition and reacquisition time. Acquisition and reacquisition time shall be 10 seconds, maximum, at any data rate, 95 percent of the time upon application of the TDM signal with a BER of $10^{-3}$, or less.

3.2.5.5 Bit count integrity. The maximum probability of losing frame synchronization due to bit errors on a per bit basis shall be $10^{-15}$ at a BER of $10^{-3}$, or less.



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Frame length - 10336 Bits

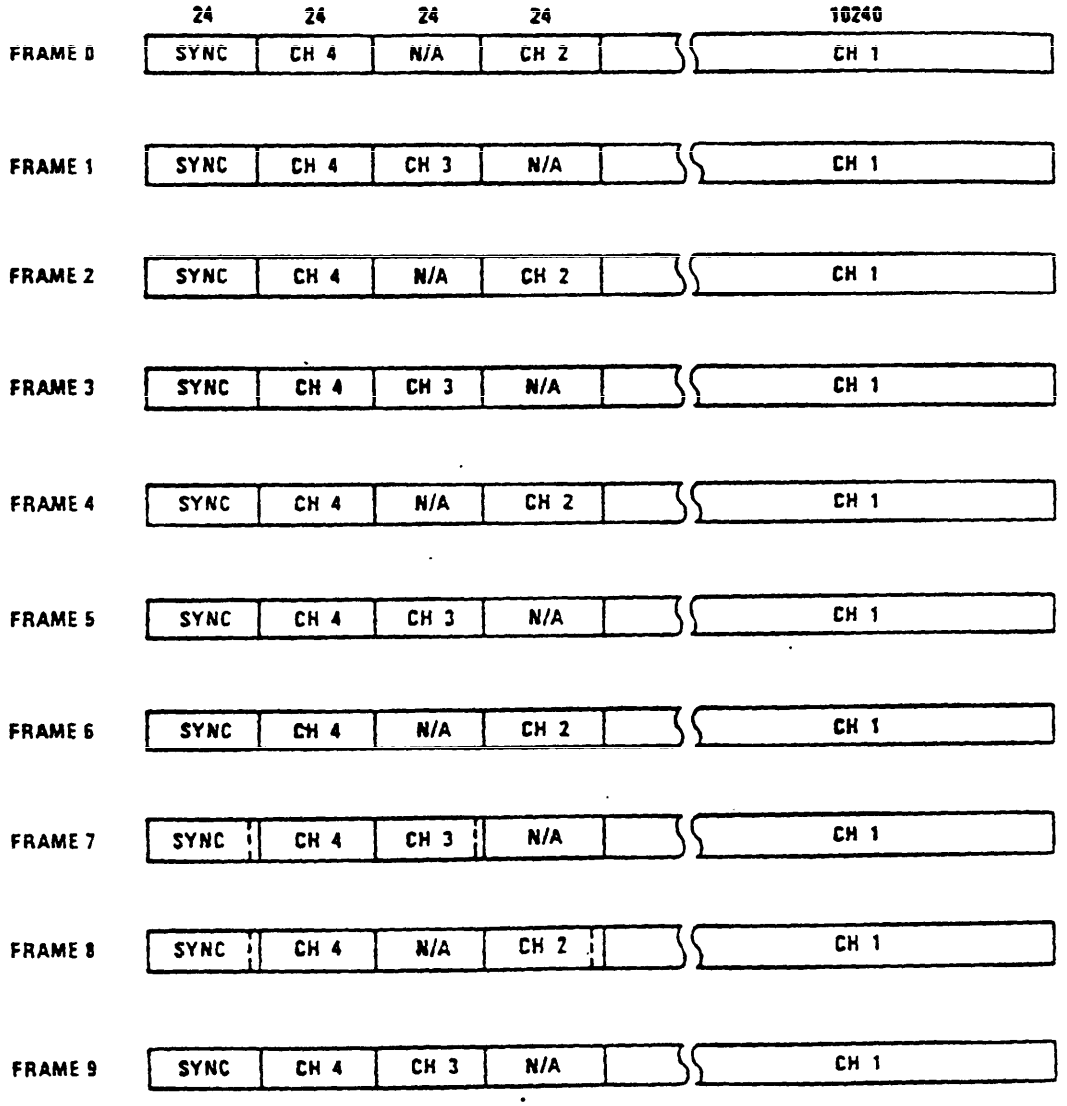
Superframe length - 51680 Bits

Transport rate - 32.3 kbps

FIGURE 9. 32 kbps-Channel Mode.



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Frame length - 10336 Bits

Superframe length - 103360 Bits

Transport rate - 64.6 kbps

FIGURE 10. 64 kbps-Channel Mode.



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3.2.5.6 Frame synchronization performance. The 24 bits allocated for sync information specified in 3.2.5.1.1 through 3.2.5.1.5 is actually only 23 bits. The twenty-fourth bit is only used to hold positive stuff bits when they occur. The first 23 bits of each frame contain one of four possible sync words depending on what type of stuffing operation, if any, is occurring in that particular frame. One sync word is only used to designate the beginning of a superframe and always occurs in Frame 0. The other three sync words represent positive stuffing, negative stuffing, and no stuffing operations. The four sync words are:

| | |
|---|---|
| Superframe | 0 0 1 1 1 1 1 1 0 1 0 0 0 0 0 0 1 0 0 0 1 0 1 |
| +Stuff | 1 1 0 0 1 1 1 0 1 0 1 1 1 1 1 1 0 1 1 0 1 0 1 |
| -Stuff | 1 1 0 0 0 0 0 1 0 1 0 1 1 1 0 0 1 0 1 1 0 1 0 |
| No stuff | 0 0 1 1 0 0 0 0 1 0 1 0 0 0 1 1 0 1 0 1 0 1 0 |

These four sync words are specifically chosen to exhibit maximum distance cross-correlation characteristics as well as desirable auto-correlation characteristics. When compared to one another, any two of the four sync words show at least 15 bit differences making it difficult to mistake one word for another even with random bit errors added. When any of the sync words is compared to itself (shifted in either direction by one or more bits), there is little comparison. These characteristics enable the DEMUX to locate and recognize each sync word with speed and reliabilty.

3.3 Physical characteristics. The modem unit shall have no protrusions greater than 1 inch (in.) in front of the panel face excluding handles which may extend up to 2 in. from the panel face. Front panel switches shall be types which minimize accidental operation. Status indicators shall be visible at all times.

3.3.1 Dimensions. The front panel of the modem shall be the standard 19-in. rack width by the standard 8.75-in. panel height. The front panel shall be 0.125-in. thick. The chassis unit shall include front panel retainers and bearing blocks. The overall depth of the chassis from the rear of the front panel shall not exceed 22.0 in., not including connectors and fan blower. FIGURE 11 is a chassis outline DWG indicating significant overall dimensions.

3.3.2 Weight. The weight of the modem unit including the chassis portion of the rack slides shall not exceed 80 pounds. Appropriate two-man lift warnings shall be prominently labeled.

3.4 Environmental conditions. The modem shall operate continuously and be designed to conform to all performance requirements of this specification when subjected to any combination of environmental conditions within the operating ranges specified herein. The modem shall be designed to conform to



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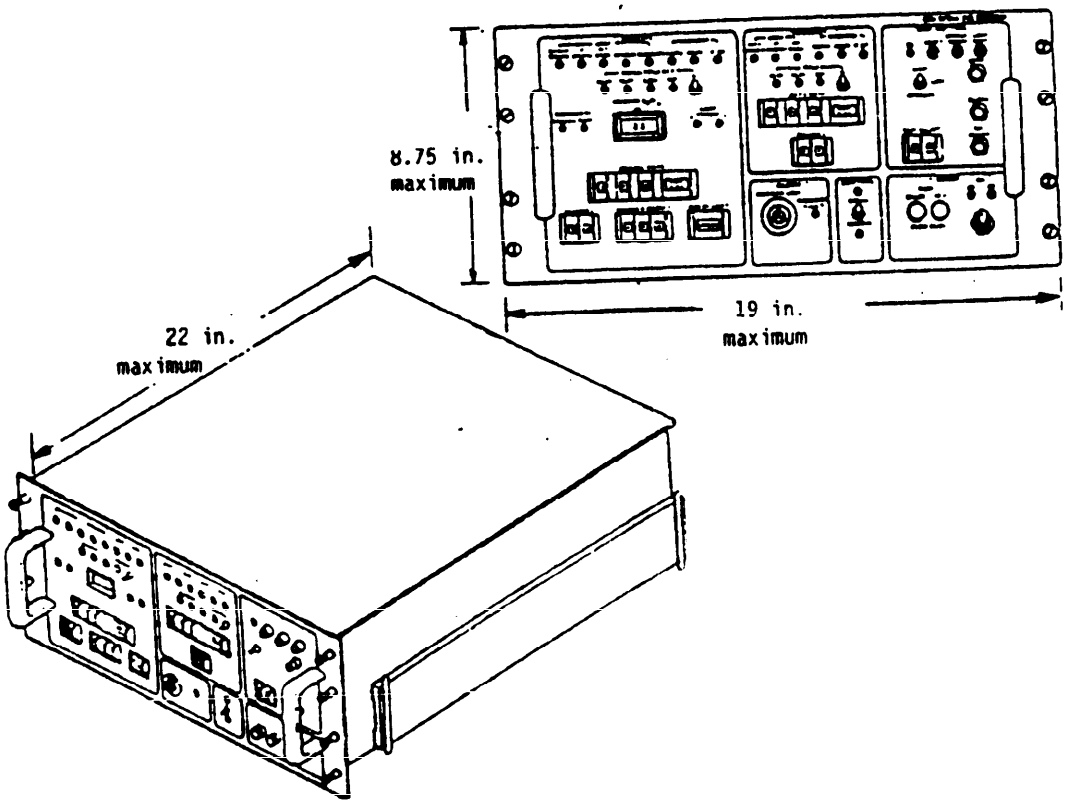



FIGURE 11. Chassis dimensions.

all performance requirements of this specification without repair after being subjected to any combination of environmental conditions within the specified nonoperating ranges:

| | Operating | Nonoperating |
|---|---|---|
| Temperature (in degrees Celsius (C)) | 0 to 50 | -28 to 65 |
| Relative humidity (without condensation) | Up to 95 percent | Up to 95 percent |
| Altitude, feet | Up to 15,000 | Up to 40,000 |



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3.4.1 Ruggedness. All modules, chassis, and drawers shall withstand shipment, bench repair, and installation handling without incurring damage to operating controls, meters, test points, and circuitry. All cabling in modules, chassis, and drawers shall be routed, dressed, and laced in accordance with MIL-STD-454, Requirement 9. The equipment shall withstand the bench handling shock in accordance with MIL-STD-810, Method 516.3, Procedure I (5gs) and Procedure VI, and Shipboard shock in accordance with MIL-S-901 for Grade A, Class I, lightweight, and vibration in accordance with MIL-STD-167-1, Type I, and 4 Hz to 50 Hz with 0.03 in. to 0.003 in. single amplitude.

3.4.2 Inclination. The modem shall be designed to withstand repeated inclinations of up to 45 degrees.

3.4.3 Magnetic field environment. The modem shall not be damaged (may be degraded during application) when subjected to direct current (DC) magnetic field environment as specified in DoD-STD-1399, Section 070, Part 1.

3.4.4 Salt spray. The modem shall withstand the effects of FED-STD-151, Method 811.1, when configured in the Transport Mode for shipping.

3.5 Parts, materials, and processes. All logic where design permits shall consist of the 74 ceramic series, or equivalent. Commercial parts may be used provided they are appropriately marked and are identified on the DWGs and parts lists for the modem by manufacturer's identification and part number.

3.5.1 Parts. Parts shall be as specified in 3.5.1.1 through 3.5.1.7.

3.5.1.1 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.5.1.2 Electrostatic discharge (ESD). Equipment built under the contract, which requires design or redesign, shall be provided electrostatic protection in accordance with the Design protection paragraph of DoD-STD-1686.

3.5.1.3 Front panel fasteners. Front panel fasteners shall be of quick-release type which shall withstand 1000 front panel openings and closings without failure to function.

3.5.1.4 Input and output (I/O) interface connectors. I/O connectors shall be as specified in TABLE VIII.

3.5.1.5 Rescreening of semiconductor, miroelectronic, and hybrid devices. All semiconductor, microelectronic, and hybrid devices shall be rescreened in accordance with the procedures specified in the manufacturer's quality program plan as approved by the Government.



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TABLE VIII. I/O interface connectors.

| Function | Signal level | Load/source impedance (ohms) | Connector type | Mating connector |
|---|---|---|---|---|
| Prime power input | 115 VAC | - | MS3124E12-3P | MS3126F12-3S |
| PN sync output | TTL | - | BNC1/ | M39012/16-0001 |
| Error CLK output | TTL | - | BNC1/ | M39012/16-0001 |
| Error output | TTL | - | BNC1/ | M39012/16-0001 |
| Carrier lock relay | Relay closure | - | TWINAX | AMPHENOL 31-224 |
| Modulator IF output | 0 dBm | 50 | TNC | M39012/26-0101 |
| Demodulator IF input | -80 dBm to -10 dBm | 50 | TNC | M390122/26-0101 |
| Modulator (Channel 1) reference clock output | MIL-STD-188-114 | 100 | M24308/2-3 M24308/4-3 (see FIGURE 12) | |
| Modulator (Channel 1)2/ clock input | MIL-STD-188-114 | >6k | | |
| Modulator (Channel 1)2/ data input | MIL-STD-188-114 | >6k | | |
| Demodulator (Channel 1)2/ clock output | MIL-STD-188-114 | 100 | | |
| Demodulator (Channel 1)2/ data output | MIL-STD-188-114 | 100 | | |
| 5-MHz reference input | 0 dBm to +13 dBm | 50 | TNC | M39012/26-0101 |
| LOCC or ROCC | 0 VDC to +5 VDC | - | MS3110E-14-19P | MS3116F14-19S |



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TABLE VIII. I/O interface connectors - Continued.

| Function | Signal level | Load/source impedance (ohms) | Connector type | Mating connector |
|---|---|---|---|---|
| | MUX/DEMUX interface connectors | | | Mating connector |
| MUX clock (Channel 1, 2, and 3) Inputs (three each) | MIL-STD-188-114 | >6k | | |
| MUX data inputs (Channel 1, 2, and 3) Inputs (three each) | MIL-STD-188-114 | >6k | M24308/2-3 (See FIGURE 12 | M24308/4-3 and FIGURE 13) |
| DEMUX clock (Channel 1, 2, and 3) Outputs (three each) | MIL-STD-188-114 | 100 | | |
| DEMUX data (Channel 1, 2, and 3) Outputs (three each) | MIL-STD-188-114 | 100 | | |
| DEMUX lock relay | Relay closure | - | TWINAX | AMPHENOL 31-224 |
| | Additional interface connectors | | | |
| Composite MUX clock (test point) | TTL | - | | |
| Composite MUX data (test point) | TTL | - | | |



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TABLE VIII. I/O interface connectors - Continued.

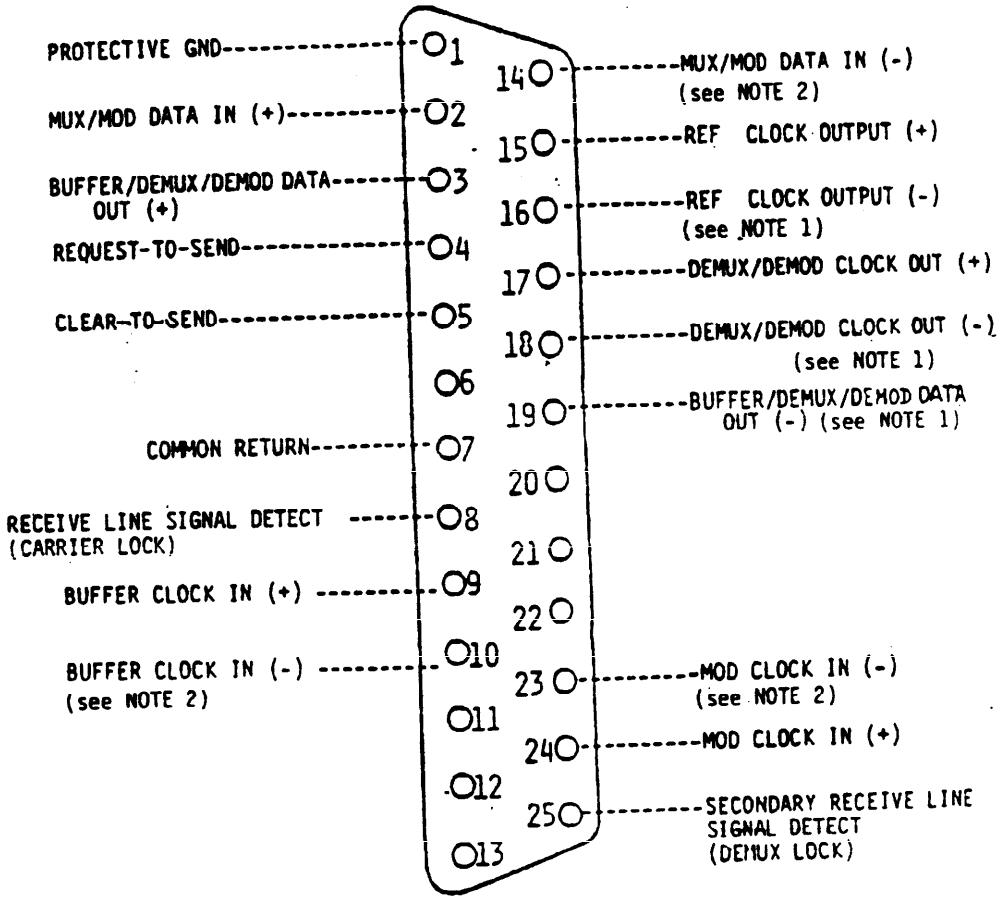
| Function | Signal level | Load/source impedance (ohms) | Connector type | Mating connector |
|---|---|---|---|---|
| Composite DEMUX clock (test point) | TTL | - | M24308/2-3 See FIGURE 14 | M24308/4-3 |
| Composite DEMUX data (test point) | TTL | - | | |
| Summary fault | MIL-STD-188-114 (unbalanced) | 100 | | |
| Request-to-send (modulator key input) | EIA RS-232-C-69 | - | | |
| Clear-to-send (modulator ON status output) | EIA RS-232-C-69 | - | M24308/2-3 See FIGURE 12 | M24308/4-3 |
| Receive line signal detect (carrier lock) | EIA RS-232-C-69 | - | | |
| Secondary receive line signal detect (DEMUX lock) | EIA RS-232-C-69 | - | | |
| Summary fault relay | Relay closure | - | TWINAX | AMPHENOL 31-224 |

1/ Front panel access

2/ Part of MUX/DEMUX I/O connectors when in the MUX Mode



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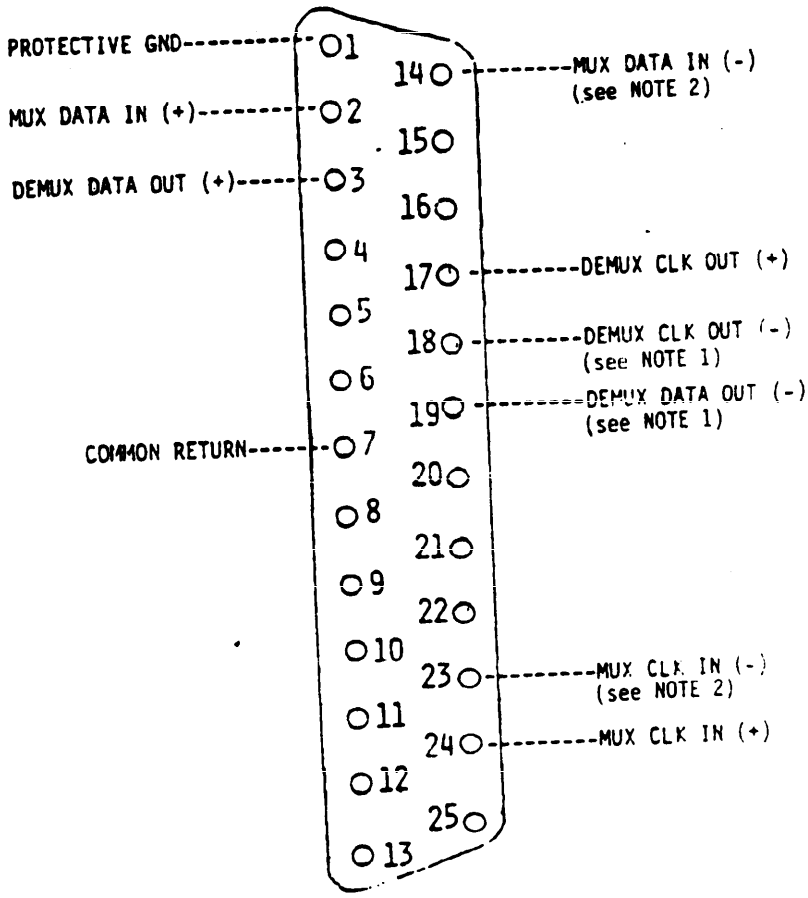



NOTES:
1. No connection is required for unbalanced operation.
2. Signal ground connection required for unbalanced operation.

FIGURE 12. Pin assignment for Channel 1 connector.



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NOTES:
1. No connection is required for unbalanced operation.
2. Signal ground connection required for unbalanced operation.

FIGURE 13. Pin assignment for Channel 2 and Channel 3 connectors.



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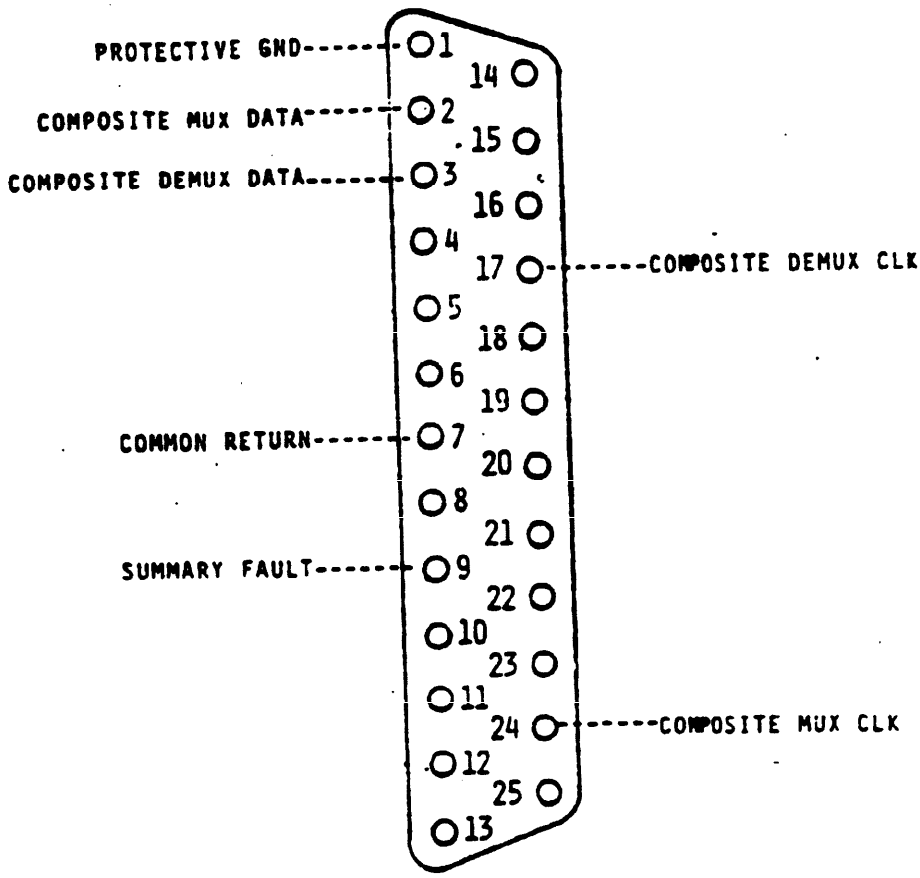



FIGURE 14. Pin assignment for test point connector.



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3.5.1.6 Printed wiring. Printed wiring shall be in accordance with MIL-STD-275 and MIL-P-55110.

3.5.1.7 Chassis interface. All interface multipin mating connectors (excluding TNC and BNC types) shall be supplied for external interfaces.

3.5.2 Materials. Materials shall be as specified in 3.5.2.1 and 3.5.2.2.

3.5.2.1 Fungus. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, such materials shall be protected against moisture and fungus with a fungus-resistant varnish. All exposed metallic conduction surfaces shall be protected against atmospheric corrosion by coating or by another method which provides equal protection.

3.5.2.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered in storage or normal service. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other as specified in the Dissimilar metals paragraph of MIL-E-16400G.

3.5.3 Processes. Processes shall be as specified in 3.5.3.1 through 3.5.3.3.

3.5.3.1 Finish. The modem shall be given a protective finish in accordance with MIL-F-14072. The exterior surfaces of the modem shall be painted with a semigloss, light gray enamel, Color No. 26307 in accordance with FED-STD-595.

3.5.3.2 Conformal coating. Conformal coating shall be applied to all printed wiring assemblies. Encapsulation or embedment processes shall not be used.

3.5.3.3 Soldering. Soldering shall be in accordance with WS-6536E, SCN 1, SCN 2, and SCN 3.

3.6 Design and construction. Design and construction shall be as specified in 3.6.1 through 3.6.3.

3.6.1 Electrical design and construction. Electrical design and construction shall be as specified in 3.6.1.1 through 3.6.1.4.

3.6.1.1 Input primary power. The modem shall be designed to operate with input primary power of single-phase, 3-wire (line neutral and safety ground) conforming to the requirements of DoD-STD-1399, Section 300, Type I, as specified in a through c:

a. Voltage, 115 volts alternating current (VAC) nominal



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b. Frequency, 60 Hertz (Hz)
c. Power dissipation, 400 watts (W) maximum alternating current (AC) average

3.6.1.2 Leakage current. The rms leakage current caused by electrical filters (capacitors) connected from phase to ground shall be limited to 30 milliamperes (mA) per phase for each unisolated input to the equipment. The leakage current shall be the total rms current (up to the frequency specified) per phase which will pass through the filters from phase to ground when a voltage waveform having the voltage and frequency characteristics of the Leakage current, phase-to-ground paragraph of MIL-E-16400G is impressed from the input phase-to-ground of the equipment.

3.6.1.3 Grounding and bonding. Grounding and bonding shall be as specified in the Grounding and bonding paragraph of MIL-E-16400G.

3.6.1.4 Power supply. Power supplies which require design or redesign shall be low voltage, delivering up to 5 kilowatts at 300 volts direct current (VDC), or less, and shall comply with the requirements specified in a through m:

a. The power density shall be output power divided by power supply envelope volume, including electromagnetic interference (EMI) filtering, where required. Power density exceeding 2 W per cubic inch shall require the approval of the procuring activity.
b. Junction temperature of semiconductor and microelectronic devices (diodes, transistors, hybrid and ICs, and so forth) shall not exceed +110°C under worst-case conditions.
c. Case or hot spot temperatures shall not exceed 40°C rise above ambient temperature with a maximum temperature of +110°C for parts less than or equal to 3 W of dissipation; 55°C rise above ambient with a maximum temperature of +125°C for parts greater than 3 W of dissipation; 30°C rise above part ambient with a maximum temperature of +100°C for transformers; and 10°C rise above ambient, due to self-heating, with a maximum temperature of 85°C for capacitors.
d. The use of nonstandard parts shall not be prohibited, provided there are at least two independent sources; requests to utilize nonstandard parts shall comply with existing Government procedures.
e. Power supplies shall be designed to be maintainable at either intermediate or depot level; the power supplies shall not be limited by design to repair by the manufacturer.
f. Power supplies shall not be encapsulated or embedded (potted) unless it can be shown by analyses and test to be necessary for heat removal or dissipation. This requirement shall not exclude conformal coating.
g. Power supply manufacturing shall include random vibration and temperature cycling of every unit, under full electrical load; this may be done in the end item but shall be performed on a unit basis for spares.



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h. Equipment incorporating power supplies shall be designed to operate from Type I power specified in DoD-STD-1399, Section 300. Power supplies shall be tested with the end equipment, or shall be tested as a unit by duplicating the input and output power of the power supply as installed for operation in the end equipment.

i. EMI requirements shall be tailored for the power supply which will enable the end equipments to conform to the class of MIL-STD-461 required by the requirements.

j. The mean-time-between-failures (MTBF) requirement shall be not less than 40,000 hours for Naval sheltered equipment, at 55°C.

k. Power supplies shall be designed to withstand and shall not be damaged by any load between an open circuit and a short circuit.

l. Power supplies shall be designed with a reserve load capacity of 20 percent.

m. Additional guidance for design and manufacturing of power supplies is contained in NAVMAT P 4855-1 and MIL-HDBK-241.

3.6.2 Mechanical design. Mechanical design shall be as specified in 3.6.2.1 through 3.6.2.4.1.

3.6.2.1 Modularity and commonality. Modular construction shall be employed and the equipment shall be designed for fixed-plant installation. The modem shall be designed with replaceable plug-in modules and plug-in printed circuit cards. PCBs used within replaceable plug-in modules are not required to be plug-in. Plug-in modules and cards shall be designed to permit maximum interchangeability. All replaceable components shall be limited to as few different types as possible to simplify spare parts stocking requirements. All plug-in modules and cards shall require no unsoldering to remove the modules or cards. Modules and cards shall be securely locked in position when in place so that they cannot be loosened by vibration. Conformance to this requirement may be achieved by mechanical detent device. Friction alone is not adequate.

3.6.2.2 Chassis slides. The modem shall contain positive lockout chassis slides for mounting in a standard 19-in. rack as specified in EIA RS-310-C-77. Recommended extension length is 22 in. Tilt slides are not required. Slides including the mating rack portion shall be provided with each unit.

3.6.2.3 Controls, monitors, and alarms. Controls, monitors, and alarms shall be as specified in 3.6.2.3.1 through 3.6.2.3.4.

3.6.2.3.1 Operating controls and monitors. The modem shall provide the necessary controls, indicators, and interface signals for normal communication as specified in TABLE I. All internal control switches shall be of the same type. A graphic aid shall be provided internal to the modem. This aid shall provide a means to note normal internal switch settings for operation in a particular modem configuration.



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3.6.2.3.2 Test controls and indicators. The modem shall provide the necessary controls, indicators, and monitor signals for link and loop-back testing. The test control and monitor functions shall be as specified in TABLE V.

3.6.2.3.3 Temperature warning alarm. The unit shall provide an over temperature warning status signal to the terminal control console.

3.6.2.3.4 Factory adjustment. Controls which do not require adjustment during routine maintenance and operation shall be sealed at the factory with glyptol, or equivalent.

3.6.2.4 Cooling. The modem may be designed to operate over the specified operating temperatures with the use of blowers.

3.6.2.4.1 Cooling air. The modem shall receive cooling air through side vents on the forward portion of the unit and exhaust air out of the rear panel. Inlet air shall be between 0°C and 50°C. An airflow of 90 cubic feet per minute at 0.1 in. $H_2O$ drop minimum is required at the input. A free air space of 2.0 in. minimum shall be provided behind the modem rear panel exhaust port.

3.6.3 Fail safe design. It shall not be possible to obtain any combination of front panel control settings which will result in damage to the modem functions or any associated equipment.

3.7 Electromagetic environmental effects ($E^3$). The modem shall be self-compatible and shall operate compatibly within its electromagnetic environment. As a minimum, the modem shall conform to the MIL-STD-461 requirements for Class A3 and Class A4 modem equipment, including RS05, with an equipment recovery time of not more than 5 seconds.

3.8 Human engineering. Human engineering shall conform to the requirements of the paragraphs of MIL-STD-1472 specified in a through c:

a. Control Display Integration
b. Visual Displays
c. Controls

3.9 Transportability. The modem shall be capable of transportation by air, sea, rail, or truck when configured in the Transport Mode for shipping.

3.10 Identification and marking. All markings shall be clear, concise, legible, and permanent. The modem, equipment, assemblies, and modules shall be marked with identification as specified in 3.10.1 through 3.10.4.3.

3.10.1 Controls. All controls shall be clearly marked in a manner which indicates their function and enhances the ability to adjust the modem.



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3.10.2 Components and modules. All lamps, switches, fuses, jacks, and test points shall be marked. All other components referenced in DWG 628601-G01 shall be marked wherever possible.

3.10.3 Wiring. All terminal strips shall have each terminal numbered. Each removable cable and receptacle shall be marked with permanent cable band or stencil, respectively, giving referenced designators.

3.10.4 ESD marking. ESD marking shall be as specified in 3.10.4.1 through 3.10.4.3. ESD marking shall apply only to the specific designed or redesigned item, not to the next higher assembly.

3.10.4.1 Assemblies. Assemblies shall be marked in accordance with DoD-STD-1686.

3.10.4.2 Equipment enclosures. Equipment enclosures shall be marked in accordance with DoD-STD-1686.

3.10.4.3 Nonstandard parts. ESD nonstandard parts shall be marked with the sensitive electronic device symbol of MIL-STD-129.

3.11 Safety. All construction and material installation performed by the contractor shall be designed to prevent injury to personnel or equipment during installation, operation, and maintenance. The personnel hazard safety shall conform to MIL-STD-454, Requirement 1.

3.12 Reliability. The modem shall have an upper test MTBF ($\theta_0$ as defined by MIL-STD-781C) of 1000 hours. The lower test MTBF ($\theta_1$ as defined by MIL-STD-781C) of the modem shall be 500 hours.

3.13 Maintainability. The mean-time-to-repair (MTTR) (as defined by MIL-STD-471) of the modem shall be no greater than 30 minutes, and shall provide an inherent availability of at least 0.999. The MTTR of the modem includes its fault definition and isolation electronics when repair is accomplished at the organizational level, or maintenance by the replacement of the failed subassembly (printed circuit card, module, and so forth). The maximum-corrective-maintenance-time ($M_{maxct}$) of the modem, at the 95th percentile, shall not exceed 1 hour. Corrective maintenance time includes that time required for localization, isolation, disassembly, interchange, reassembly, alignment, and checkout. The modem is considered in a corrective action mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.13.1 Organizational level maintenance. The modem shall be designed to require minimum logistic support at the organization (user) level. The unit shall consist of modular assemblies. These assemblies shall be PCBs which slide into card guides. The chassis shall contain subassemblies consisting of power supplies and a fan. These subassemblies shall be accessible by removing the unit top cover.



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3.13.2 Depot level maintenance. The modem shall be basically a digital design which will lend itself to automatic testing and fault isolation to the component level.

3.13.3 Built-in test equipment (BITE) preventive maintenance and fault location. The modem shall have integral test facilities for the measurements, testing, and monitoring of signals in and out of the modem and of the operation of the internal circuits. The BITE shall not require auxiliary test equipment such as oscilloscopes or transmission measuring sets for normal system operation. The integral test facilities shall include a data source, a method to monitor DC power supply voltages, and GO-NO-GO fault locator circuitry to isolate failures to a particular plug-in card or group of cards. Metering facilities or other indicating devices in conjunction with switches shall be provided to ensure proper operation of the modem and to monitor the DC voltages, GO-NO-GO fault location function and other critical internal and external signals. The test signal shall be compatible with the HP-1645A BER test set 511 bit ($2^9$ - 1) test sequence.

3.14 Workmanship. Workmanship shall be as specified in 3.14.1 and 3.14.2.

3.14.1 General workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.14.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration and temperature cycling as specified in 4.8.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subjected to Government verification at any time. Verification will consist of, but not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of packaging (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IX.

TABLE IX. Examinations and tests.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Digital data input interface | 3.2.1.1 | 4.5 | X | X | | |
| Mode 1 operation | 3.2.1.1.1 | 4.5 | X | X | | |
| Mode 2 operation | 3.2.1.1.2 | 4.5 | X | X | | |
| Input data rate range | 3.2.1.2 | 4.5 | X | X | | |
| Output frequency | 3.2.1.3 | 4.5 | X | X | | |
| Output level | 3.2.1.4 | 4.5 | X | X | | |
| Output impedance | 3.2.1.5 | 4.5 | X | X | | |



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TABLE IX. Examinations and tests - Continued.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Output modulation | 3.2.1.6 | 4.5 | X | X | | |
| Antipodal keying error | 3.2.1.7 | 4.5 | X | X | | |
| AM of signal component | 3.2.1.8 | 4.5 | X | X | | |
| Output spectrum | 3.2.1.9 | 4.5 | X | X | | |
| Encoding | 3.2.1.10 | 4.5 | X | X | | |
| Encoding switch | 3.2.1.10.1 | 4.5 | X | X | | |
| One-half rate evolutional coding | 3.2.1.10.2 | 4.5 | X | X | | |
| Output clock | 3.2.1.11 | 4.5 | X | X | | |
| Input frequency | 3.2.2.1 | 4.5 | X | X | | |
| Input impedance | 3.2.2.2 | 4.5 | X | X | | |
| Input modulation | 3.2.2.3 | 4.5 | X | X | | |
| Antipodal keying error | 3.2.2.4 | 4.5 | X | X | | |
| AM of signal component | 3.2.2.5 | 4.5 | X | X | | |
| Input noise spectral density | 3.2.2.6 | 4.5 | X | X | | |
| Minimum input signal power | 3.2.2.7 | 4.5 | X | X | | |
| Maximum total IF input power | 3.2.2.8 | 4.5 | X | X | | |
| Nondamage input power level | 3.2.2.9 | 4.5 | X | X | | |
| Image rejection | 3.2.2.10 | 4.5 | X | X | | |
| Signal acquisition range | 3.2.2.11 | 4.5 | X | X | | |
| Signal acquisition time | 3.2.2.12 | 4.5 | X | X | | |
| Carrier lock threshold | 3.2.2.13 | 4.5 | X | X | | |
| Acquisition range | 3.2.2.14.1 | 4.5 | X | X | | |
| Acquisition time | 3.2.2.14.2 | 4.5 | X | X | | |
| Bit timing loop input transition density | 3.2.2.15 | 4.5 | X | X | | |
| Bit timing loop input isochronous distortion | 3.2.2.16 | 4.5 | X | X | | |
| Bit count integrity | 3.2.2.17 | 4.5 | X | X | | |
| Output data | 3.2.2.18 | 4.5 | X | X | | |
| Output clock | 3.2.2.19 | 4.5 | X | X | | |
| Rms output timing jitter | 3.2.2.20 | 4.5 | X | X | | |



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TABLE IX. Examinations and tests - Continued.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Output data rate range | 3.2.2.21 | 4.5 | X | X | | |
| Decoding | 3.2.2.22 | 4.5 | X | X | | |
| Coding gain | 3.2.2.23 | 4.5 | X | X | | |
| Demodulator bit synchronizer and decoder performance | 3.2.2.24 | 4.5 | X | X | | |
| Performance with applied distortion | 3.2.2.25 | 4.5 | X | X | | |
| Doppler acceleration performance | 3.2.2.25.1 | 4.5 | X | X | | |
| Adjacent channel interference performance | 3.2.2.25.2 | 4.5 | X | X | | |
| Data buffering | 3.2.2.25.3 | 4.5 | X | X | | |
| Remote status | 3.2.3.1 | 4.5 | X | X | | |
| Remote control | 3.2.3.2 | 4.5 | X | X | | |
| MUX | 3.2.4 | 4.5 | X | X | | |
| Input data rates | 3.2.4.1 | 4.5 | X | X | | |
| Input data rate accuracy | 3.2.4.2 | 4.5 | X | X | | |
| Data clock interface | 3.2.4.3 | 4.5 | X | X | | |
| Composite transmit rates | 3.2.4.4 | 4.5 | X | X | | |
| DEMUX | 3.2.5 | 4.5 | X | X | | |
| Modes/output rates | 3.2.5.1 | 4.5 | X | X | | |
| Seventeen-Channel Mode | 3.2.5.1.1 | 4.5 | X | X | | |
| Channel input rate acceleration | 3.2.5.1.1.1 | 4.5 | X | X | | |
| Stuff-bit operation | 3.2.5.1.1.2 | 4.5 | X | X | | |
| Seven-Channel Mode | 3.2.5.1.2 | 4.5 | X | X | | |
| Three-Channel Mode | 3.2.5.1.3 | 4.5 | X | X | | |
| 32 kbps-Channel Mode | 3.2.5.1.4 | 4.5 | X | X | | |
| 64 kbps-Channel Mode | 3.2.5.1.5 | 4.5 | X | X | | |
| Output data rate accuracy | 3.2.5.2 | 4.5 | X | X | | |
| Data and clock interface | 3.2.5.3 | 4.5 | X | X | | |
| Acquisition and reacquisition time | 3.2.5.4 | 4.5 | X | X | | |



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TABLE IX. Examinations and tests - Continued.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Bit count integrity | 3.2.5.5 | 4.5 | X | X | | |
| Frame synchronization performance | 3.2.5.6 | 4.5 | X | X | | |
| Physical characteristics | 3.3 | 4.5 | X | | X | |
| Dimensions | 3.3.1 | 4.5 | X | | | |
| Weight | 3.3.2 | 4.5 | X | | | |
| Environmental conditions | 3.4 | 4.5 | X | | | X |
| Ruggedness | 3.4.1 | 4.5 | X | | | X |
| Inclination | 3.4.2 | 4.5 | X | | | X |
| Magnetic field environment | 3.4.3 | 4.5 | X | | | X |
| Salt spray | 3.4.4 | 4.5 | X | | | X |
| Parts, materials, and processes | 3.5 | 4.5 | X | | X | |
| Interchangeability | 3.5.1.1 | 4.5 | X | | | |
| ESD | 3.5.1.2 | 4.5 | X | | X | |
| Front panel fasteners | 3.5.1.3 | 4.5 | X | X | | |
| I/O interface connectors | 3.5.1.4 | 4.5 | X | X | | |
| Rescreening of semiconductor, microelectronic, and hybrid devices | 3.5.1.5 | 4.5 | X | | X | |
| | 3.5.1.6 | 4.5 | X | | X | |
| Printed wiring | 3.5.1.7 | 4.5 | X | | X | |
| Chassis interface | 3.5.2.1 | 4.5 | X | | X | |
| Fungus | 3.5.2.2 | 4.5 | X | | X | |
| Metals | 3.5.3.1 | 4.5 | X | X | X | |
| Finish | 3.5.3.2 | 4.5 | X | | X | |
| Conformal coating | 3.5.3.3 | 4.5 | X | | X | |
| Soldering | 3.6.1.1 | 4.5.2 | X | X | X | |
| Input primary power | 3.6.1.2 | 4.5.2 | X | | X | |
| Leakage current | 3.6.1.3 | 4.5 | X | | X | |
| Grounding and bonding | 3.6.1.4 | 4.5.2 | X | | X | |
| Power supply | | | | | | |
| Modularity and commonality | 3.6.2.1 | 4.5 | X | | | |
| Chassis slides | 3.6.2.2 | 4.5 | X | | X | |



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TABLE IX. Examinations and tests - Continued.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection | | |
|---|---|---|---|---|---|---|
| | | | | Group A | Group B | Group C |
| Operating controls and monitors | 3.6.2.3.1 | 4.5 | X | X | | |
| Test controls and indicators | 3.6.2.3.2 | 4.5 | X | X | | |
| Temperature warning alarm | 3.6.2.3.3 | 4.5 | X | X | | |
| Factory adjustment | 3.6.2.3.4 | 4.5 | X | | | |
| Cooling | 3.6.2.4 | 4.5 | X | | | |
| Cooling air | 3.6.2.4.1 | 4.5 | X | | | |
| Fail safe design | 3.6.3 | 4.5 | X | | | |
| E3 | 3.7 | 4.5, 4.5.3 | X | X | | |
| Human engineering | 3.8 | 45 | X | | | |
| Transportability | 3.9 | 4.5 | X | | | |
| Identification and marking | 3.10 | 4.5 | X | | X | |
| Controls | 3.10.1 | 4.5 | X | X | | |
| Components and modules | 3.10.2 | 4.5 | X | | X | |
| Wiring | 3.10.3 | 4.5 | X | | X | |
| Assemblies | 3.10.4.1 | 4.5 | X | | | |
| Equipment enclosures | 3.10.4.2 | 4.5 | X | | | |
| Nonstandard parts | 3.10.4.3 | 4.5 | X | | | |
| Safety | 3.11 | 4.5 | X | | | |
| Reliability | 3.12 | 4.6 | X | | | |
| Maintainability | 3.13 | 4.7 | X | | | |
| Organizational level maintenance | 3.13.1 | 4.7 | X | | | |
| Depot level maintenance | 3.13.2 | 4.7 | X | | | |
| BITE preventive maintenance and fault location | 3.13.3 | 4.5 | X | X | | |
| General workmanship | 3.14.1 | 4.8.1 | X | | X | |
| Workmanship screen | 3.14.2 | 4.8.2 | X | X | | |
| Preservation, packaging, packing, and marking | 5.1 | 4.9 | X | | X | |



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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each modem offered for delivery. Production inspection shall encompass examinations and tests, which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments, and tests which detect deviations from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests shown in Group A of TABLE IX.

4.4.2 Production control inspection (Group B). Production control inspection shall include the examinations and tests specified in TABLE IX and the inspection procedures of MIL-STD-105 using special inspection level S-3 for normal, tightened, and reduced inspection. The maximum number of sample test units shall be two each. Lot acceptance criteria shall be based on a single sampling plan with a zero acceptance number. Production control inspection shall be performed on equipments that have passed production inspection. The equipments shall satisfactorily conform to the requirements of production control inspection prior to releasing the lot for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a production unit mutually agreed to by the contractor and the Government. Environmental inspection shall encompass environmental tests to prove the durability of materials, parts, units, and the equipment as a whole; life tests, simulated service tests; tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests shown in Group C of TABLE IX.

4.4.3.1 Noncompliance. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control or environmental inspections, or both, may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all defects.



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4.5 Test methods. Conformance to the requirements specified in TABLE IX shall be verified by contractor-developed test procedures, as approved by the Government, and as specified in 4.5.1 through 4.5.3.

4.5.1 Performance tests. At least daily, performance measurements shall be made to verify the performance characteristics specified in a through c:

a. Gain and sensitivity
b. Synthesizer lock
c. Frequency stability

4.5.2 Power interruption and spike tests. Power interruption tests shall be conducted in accordance with the Power interruption tests paragraph of MIL-E-16400G. The 2500 volts (V) spike test shall be performed in accordance with the requirements of DoD-STD-1399, Section 300. Source impedance for the spike shall be 25 ohms, resistive. Spikes shall be delivered line-line and line-ground for each primary power line. For each test, a positive-going spike shall be applied at the positive crest of the supply line voltage waveform and at the zero axis crossing, and a negative-going spike shall be applied at the negative crest.

4.5.3 EMI tests. Conformance to the requirements of 3.7 shall be verified by the test methods of MIL-STD-462, as specified in the Government-approved EMI test plan. These tests shall be verified during first article inspection.

4.6 Reliability qualification test. A reliability qualification test should be performed in accordance with Test Plan IVC of MIL-STD-781C on the first production unit. Environmental conditions and operational duty cycles for reliability qualification testing shall be as specified herein.

4.6.1 Failure. After the initial start of the test, any parameter deviation which requires adjustment or alignment of the equipment under test shall be classified as a failure (see 6.5).

4.6.2 Environmental conditions for reliability testing. The environmental conditions specified in a through c shall be maintained throughout the reliability qualification test:

a. Ambient temperature shall be +50°C, +0°C, -30°C
b. Vibration shall be 1g ±10 percent peak acceleration value at any nonresonant frequency between 20 Hz and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
c. Input voltage shall be 115 VAC, 60 Hz or 400 Hz. Input voltage shall be cycled in accordance with the 3-day cycle specified in 1 through 3:
   1. Nominal
   2. Nominal +10 percent
   3. Nominal -10 percent



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Signal level Impedance (ohms) Type

4.6.3 Test frequencies. Test frequencies of 69.0 MHz, 70 MHz, and 71.0 MHz shall be used during reliability testing.

4.7 Maintainabilty demonstration. Maintainability demonstration shall be as specified in 4.7.1 through 4.7.7.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be used to verify conformance to the MTTR requirements. The maintainability demonstration shall be performed just once, during the first 3 month's production.

4.7.1.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance faults shall be made available for each candidate task as specified in a through c:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative, shall use the candidate tasks as a guide to determine a sample of 50 tasks for the demonstration.

4.7.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.7.2.1.

4.7.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times shall be as specified in TABLE X.

TABLE X. Acceptance criteria.

| | MTTR | $M_{maxct}$ |
|---|---|---|
| Acceptance level<br>Sample size | 2<br>5 | 0<br>5 |



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Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index, MTTR or $M_{maxct}$, is less than or equal to that specified in TABLE X corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or less than each index. The actual value shall also be determined. An accept decision can only be made when an accept decision is made for both MTTR and $M_{maxct}$. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

4.7.3 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.7.4 Technicians. The Government reserves the right to provide and select technicians to perform the maintainability demonstration.

4.7.5 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.6 Rejection. Failure to conform to any of the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies and

b. Upon approval of an approach, repeat the demonstration until an accept decision is reached.

4.7.7 Acceptance. The maintainability demonstration shall demonstrate that the equipment conforms to the accept criteria prior to delivery.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.2.

4.8.1 General workmanship and soldering. The modem shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.5.3.3 and 3.14.1.

4.8.2 Workmanship screen. Workmanship screen shall be as specified in 4.8.2.1 and 4.8.2.2.



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4.8.2.1 Vibration. The vibration shall be performed at the chassis level. The vibration shall be random for an accumulated time of 10 minutes in the β axis. The modem shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 15. Input vibration levels shall be measured at the modem mounting points. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. Any severe structural resonances encountered shall be controlled by the use of appropriate notch filters. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed. Spare card assemblies shall be vibrated unpowered and tested separately. Failed units shall be repaired and recycled.

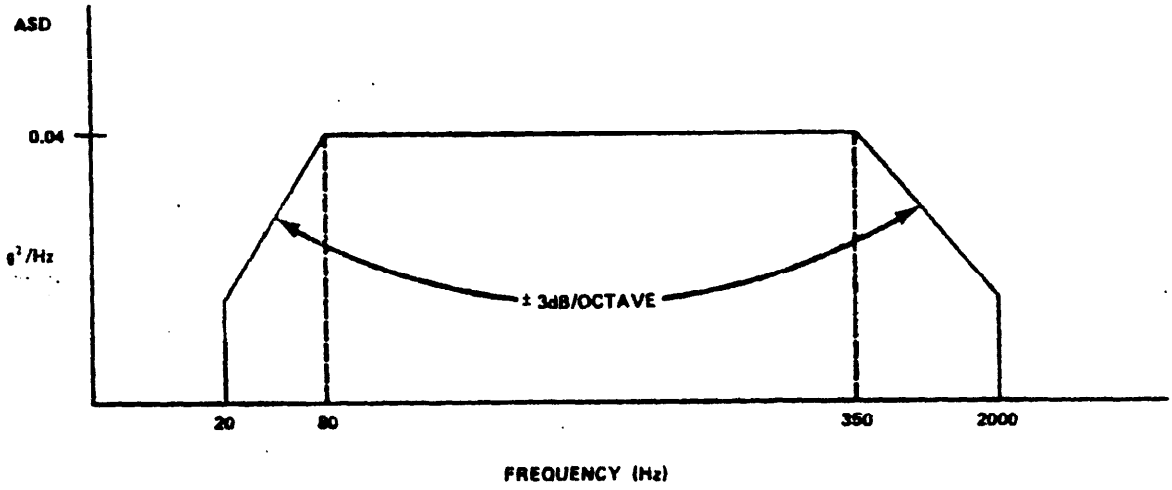



FIGURE 15. Random vibration curve.



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4.8.2.2 Temperature cycling. Each unit shall be subjected to five cycles of the temperature curve shown in FIGURE 16. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit comes within 10°C of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

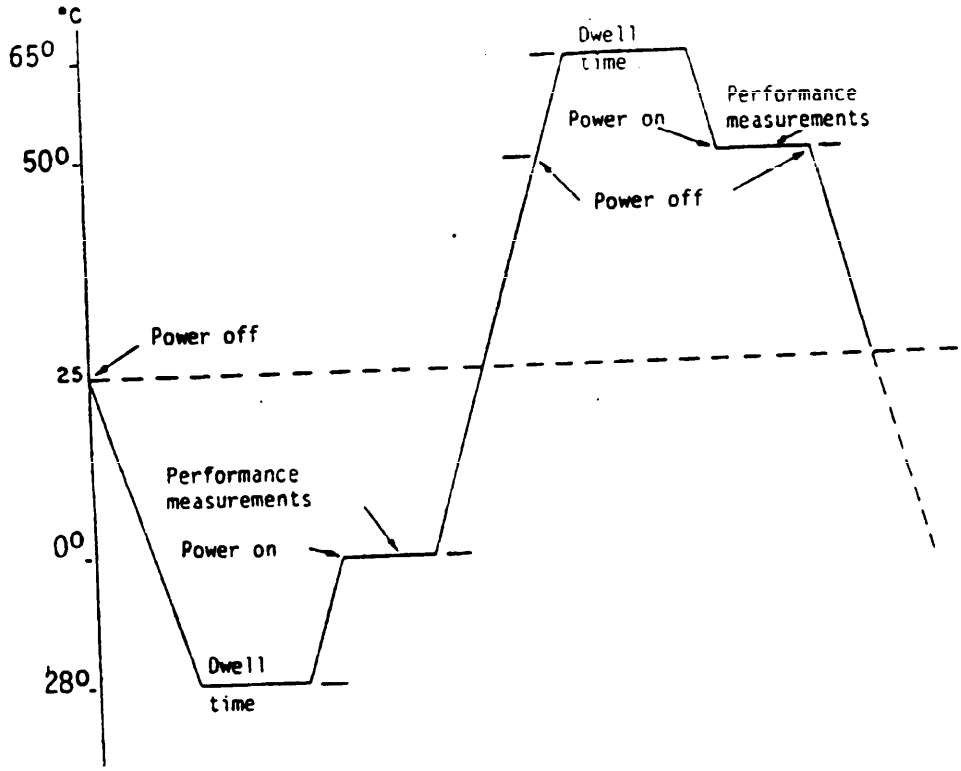



FIGURE 16. One cycle of temperature curve (not to scale).



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4.9 Inspection of packaging. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The modem covered by this specification is intended for data communications over super high frequency satellite terminal equipment.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Interchangeability requirements, if required (see 3.5.1.1)
d. Rack mounting brackets, as required
e. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)
f. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.3 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.



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6.5 Failure. Failure is defined as any malfunction or parameter deviation that prevents equipment from performing within the specified operational requirements.

6.6 Subject term (key word) listing.

Intended use: data communications over super high frequency satellite terminal equipment
MD-1030A(V), Modem, Digital Data
Modem, Digital Data, MD-1030A(V)
Satellite communications system

Preparing activity:
NAVY - EC

(Project No. 5820-N859(EC))



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